

1. Abstract

Protein aggregation limits recombinant protein production, motivating the development of short solubility-enhancing and protective peptides. We combined computational library design with experimental selection to explore variants derived from intrinsically disordered NEXT and SSB tags and the CAHS1 protective protein. NEXT and SSB libraries were evaluated using carbenicillin selection linked to aggregation-prone β -lactamase while CAHS1 variants underwent freeze-thaw selection. Sequencing of variants revealed changes in variant composition across all three libraries, supported by Monte Carlo-calibrated chi-squared tests. Property based clustering associated NEXT and SSB enrichment with greater predicted charge and hydrophilicity. There is also an enrichment of CAHS1 variants with a stronger amphiphilic helical character. These findings identify candidate property trends and sequences for further investigation. Further work is required to characterise reproducibility of findings and quantify protein solubility. This workflow provides a foundation for subsequent evolution and validation of IDP-derived solubility enhancers and protective peptides.

2. Introduction

Protein aggregation limits the production and usage of recombinant proteins by limiting soluble yield and compromising biological activity. In *Escherichia coli*, rapid protein expression can outpace productive folding, promoting rapid accumulation of insoluble inclusion bodies (Francis and Page, 2010). Addressing this problem is important in protein production for biochemical research, biotechnological application, and therapeutic development.

Solubility-enhancing protein partners such as maltose-binding proteins (MBP) and small ubiquitin-like modifiers are widely used to improve recombinant expression efficiency. However, their effectiveness varies between different target proteins, and soluble expression does not necessarily indicate correct folding and retained activity. Large tags may also introduce additional metabolic burdens, and protein applications requiring tag removal may introduce additional tag cleavage and removal steps (Jo, 2022; Tang et al., 2024). These limitations necessitate the development of shorter fusion solubility tags that improve soluble protein recovery with minimal effect to target activity.

Intrinsically disordered proteins (IDP) offer a potential source for such tags due to their disordered conformation at physiological conditions instead of a single stable structure. Flexible, solvent-exposed chains can act as entropic bristles, restricting intermolecular interactions that lead to protein aggregation (Santner et al., 2012). This principle is supported by the 52 amino acid NEXT tag which enhances soluble expression while preserving activity in fusion partners (Jo, 2022). These findings establish IDPs' potentials as solubility tags, though their performance and effect on activity remain dependent on sequence and target context.

IDP derived sequences may also support protein stability under environmental stress. Tardigrade cytosolic abundant heat-soluble (CAHS) proteins are a particularly relevant example, as they have been implicated in protecting proteins from aggregation under extreme conditions. In particular, the CAHS Core (residues 121-183) from *Paramacrobiotus richtersi* CAHS1 is found to form an amphiphilic α -helix that reduces the aggregation of proteins such as lactate dehydrogenase during dehydration-rehydration and freeze-thaw stress (Kang et al., 2024). This highlights CAHS-derived sequences as potential candidates for engineering protein stability under environmental stress.

Despite such precedents, identifying short sequences with favorable solubility and protective properties remain challenging, as conformational flexibility of IDPs complicates the modelling of properties from sequences alone. Directed evolution thus offers a way to explore sequence variation through selection through linkage to a favorable phenotype. Here, we combined computational sequence generation with experimental selection to investigate candidates derived from NEXT, SSB, and CAHS1 sequences. A De Novo library was also constructed but excluded from analysis due to contamination. For mutant libraries generated from NEXT, SSB, and De Novo sequences, selection was performed by coupling the expression of an aggregation prone mutant of β -lactamase to bacterial survival. Separately, freeze-thaw selection assessed the cellular protection associated with CAHS1. Together, these approaches aim to identify promising variants for further investigation as small solubility enhancing or protective tags, with the retention of target activity and transferability across different target proteins being longer term objectives.

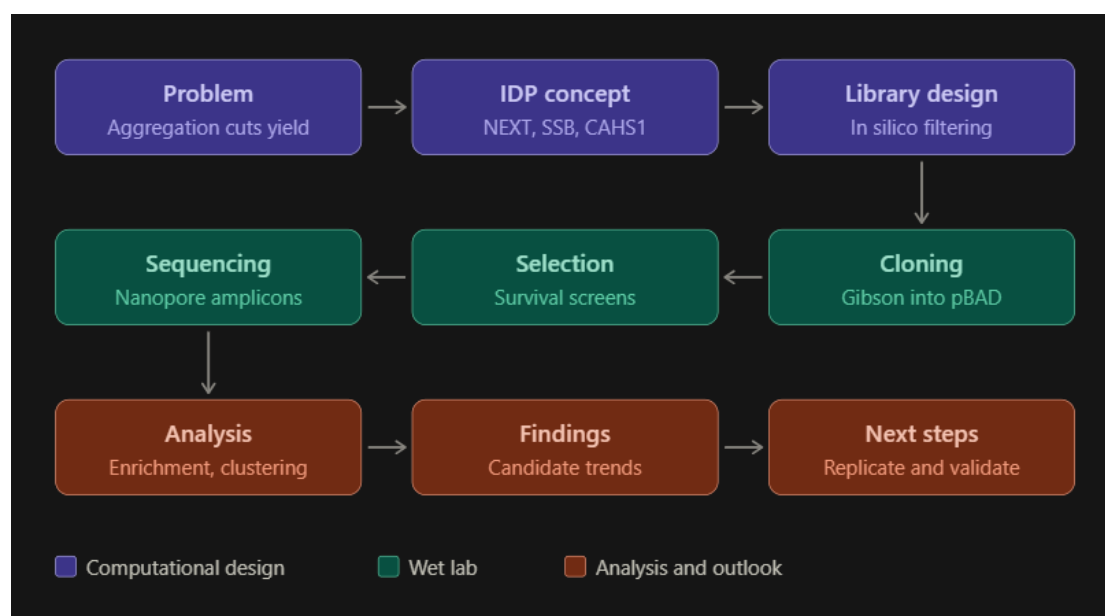


Figure 2-1. Flowchart summary of the overall project. Diagram generated by Claude Sonnet 5.5.

3. Methodology

3.1 Selecting intrinsically disordered peptide (IDP) as solubility tag

3.1.1 SSB

The single strand DNA binding protein from *E. coli* (SSB) has a 42-amino acid C-terminal peptide that lacks secondary structure and spreads laterally away from protein core (Savvides et al., 2004; Tan et al., 2017). Hydrodynamic studies and atomistic simulations also indicate that SSB C-terminal peptide form various globular conformations (Kozlov et al., 2015). Furthermore, SSB protein with C-terminal peptide deleted has increased aggregation in inclusion bodies when overexpressed (Kinebuchi et al., 1997). These suggested that SSB C-terminal peptide may sweep out area where large protein molecules are excluded, therefore reducing protein-protein aggregation via the entropic bristle mechanism. Hence, SSB was selected as a solubilizing tag for directed evolution to further enhance its performance in preventing protein aggregation.

3.1.2 NEXT

The α -carbonic anhydrase from *Hydrogenovibrio marinus* has a 52-amino acid N-terminal peptide. Sequence based predictions and circular dichroism measurements showed that this peptide is intrinsically disordered and can act as an entropic bristle, limiting intermolecular interactions and preventing aggregation. Experiments also demonstrated significant improvements in protein solubility in several recombinant proteins, including carbonic anhydrase derived from *Thermofibrio ammonificans* with minimal effect on enzyme activity (Jo, 2022). Hence, the NEXT tag was selected as a solubilising tag for directed evolution to further enhance its performance in preventing protein aggregation.

3.1.3 De Novo

Attempts on artificially constructing solubility-enhancing IDP tags (Tang et al., 2024) also inspired us to design our De Novo IDP tags. First, the composition and performance of nature-derived IDP tags (such as NEXT) and nature-derived proteins with similar properties (such as ubiquitin-like proteins and dehydrins) were analysed and were compared against that of documented artificial synthetic IDP tags. This then led to the identification of variables that potentially alter the performance of a De Novo tag: length, presence of hydrophobic amino acids, charge density, and charge distribution.

3.1.4 CAHS (not covalently linked to client protein)

The cytosolic-abundant heat soluble protein (CAHS) from the tardigrade *P. richtersi* is another intrinsically disordered protein that prevents protein aggregation and retains protein functions when cytosolic proteins are effectively very concentrated under desiccation or freeze-thaw conditions. However, instead of employing the entropic bristle mechanism, the CAHS has a core region (residues 121-183) that associates with client proteins to block client proteins from aggregating with each other (Kang et al., 2024). Within the core region, motif 1 (residues 124-142) is most critical for preventing client protein aggregation (Kang et al., 2024). Hence, the CAHS core region was selected as a solubilising protein (instead of a tag covalently linked to client protein) for directed evolution, with an emphasis on motif 1, to further enhance its performance in preventing protein aggregation.

3.2 Computational generation of semi-random mutant library

3.2.1 In silico mutagenesis

As mentioned above, the IDPs we choose to evolve use mainly overall properties to enhance client protein solubility. For instance, the entropic bristle mechanism relies on the whole peptide lacking secondary structures and flexible in conformation, so there is no single most critical residue to mutate for improving its function. By contrast, traditional directed evolution on enzymes usually focused on changing one or few catalytic residues, such as concentrating on mutating only two haem domain residues A78 and F263 to significantly improve enantioselectivity of C-H amination (Prier et al., 2017).

Therefore, large-scale mutations were applied to the amino acid sequence of each IDP. Using our self-made software ProteinAnalyzer (<https://proteinanalyzer-tfdxptasphzhddkxctm3ed.streamlit.app/>), in-silico random mutations were first applied to groups of 5 amino acids (i.e. residues 1-5, 6-10, 11-15, etc.), and 1000 mutants were generated for each group. Subsequently, similar random mutations were applied to groups of 3 amino acids (i.e., residues 1-3, 4-6, 7-9, etc.), as mutating as many as 5 residues at once may be more deleterious. For instance, changing 5 consecutive residues in an entropic bristle IDP may introduce short secondary structures like an alpha helix that tamper with the intrinsic disorder of the tag. The collection of all mutants generated becomes the preliminary library that would first pass through in-silico characterisation and selection.

As an exception, the De Novo IDP library cannot be directly generated from mutating an original sequence. Hence, our library of De Novo IDP was constructed by repeating or

recombining specific motifs while varying length, presence of hydrophobic amino acids, charge density, and charge distribution, as mentioned in Section 3.1.3. To first generate the specific motifs, amino acids were first classified into 5 groups as shown in Table 3-1. Then, various motifs at lengths of 4, 5, or 10 amino acids were generated based on the previous analysis. For instance, by analyzing composition in Smt3 protein which was used as solubility-enhancing tag, Smt3_based_motif1 was generated to resemble the positive-proline-negative-nonpolar pattern present in Smt3 sequence. Table 3-2 lists out all profiles of our designed motifs.

Subsequently, these motifs were repeated to generate De Novo IDP tags that has length of 15 ± 3 , 50 ± 3 , 100 ± 3 , 150 ± 3 , 200 ± 3 and 250 ± 3 amino acids. More De Novo IDP tags were generated by combining different types of motifs in one tag, inserting some hydrophobic patches (2 consecutive non-polar amino acids) randomly between motifs, or adding hydrophobic patch between all motifs. 3,747,989 variants of De Novo tags were generated in total as the De Novo IDP library.

Table 3-1. Categories of amino acids used in De Novo IDP library construction. *: Amino acid not favorable for motifs and would appear less frequently in De Novo library.

Category	Amino acid
Positively Charged	K, R, H*
Negatively Charged	E, D
Non-polar	G, P (ratios defined as 1:7)
Non-polar (Can be in hydrophobic patch)	G, A, V*, I*, L*
Polar	S, T, Q, N*

Table 3-2. Profiles of designed De Novo IDP motifs.

Motif name	Length	Pattern
Smt3_based_motif1	4	Positive-Proline-Negative-Nonpolar.
EB_based_motif1	5	Negative-Negative-Proline-Polar-Polar
EB_based_motif2	5	Negative-Negative-Proline-Positive-Positive
EB_based_motif3	5	Positive-Positive-Proline-Polar-Polar
EB_based_motif4	5	Polar-Polar-Proline-Negative-Negative
EB_based_motif5	5	Polar-Polar-Proline-Polar-Polar
EB_based_motif6	5	(Polar or Positive or Negative)-(Polar or Positive or Negative)-Proline-(Polar or Positive or Negative)-(Polar or Positive or Negative)
Other_rules_based_motif1	5	Have ≥ 1 nonpolar residues. Remaining residues are (Polar or Positive or Negative).
Other_rules_based_motif2	10	Have ≥ 1 nonpolar residues. Remaining residues are (Polar or Positive or Negative).

3.2.2 Characterising and in-silico selection of mutants

For all parameters mentioned below, the specific programs used to calculate each parameter can be found in https://github.com/TKA0329/iDEC_Cambridge_2026.

3.2.2.1 SSB

Since SSB was assumed to act as entropic bristle IDP, it would prevent client protein-protein aggregation by randomly changing conformation and sweeping out large spaces that exclude large molecules (Jo, 2024).

Therefore, GRAVY score was chosen as a lower GRAVY score suggests more hydrophilic residues in IDPs and increases permanent dipole-permanent dipole forces or hydrogen bonding between residues and water molecules. Furthermore, the charge on IDP at pH 7 was chosen as higher magnitude of charge increases electrostatic repulsion between client proteins attached with the same IDP.

Besides, the proportion of all secondary structures that are completely ordered (helix sheet sum) was chosen because higher proportion of such structures reduce the conformation flexibility of entropic bristle IDPs and would deteriorate its sweeping ability. Similarly, fraction of turns in secondary structure was chosen as these turns are largely disordered and higher fraction indicates more flexibility.

Finally, instability index was also chosen. This is because the mutant IDP would extend out from target protein and sweep, which would expose itself to degradation machinery in cells. If

the IDP sequence contains degradation-susceptible regions, the client protein would be degraded faster and its yield cannot accumulate.

Hence, SSB mutants in the library were characterised with these chosen parameters. Then, the mutants were compared against the parameter values from original SSB sequence. Mutants with one or more parameters that shows less benefits in solubilizing protein compared to original SSB were discarded. Table 3-3 lists out the exact in-silico mutant selection criteria for each parameter chosen here.

Table 3-3. In silico selection criteria of SSB mutants.

Parameter name	SSB Criteria
GRAVY	≤ -1.3048
Charge at pH 7	≤ -1.0 or ≥ 2.0
Helix sheet sum	≤ 0.1428
Turn fraction	≥ 0.5714
Instability index	≤ 102.6

Overall, 22000 SSB mutant sequences in library were narrowed to 132 sequences after in-silico selection.

3.2.2.2 NEXT

As NEXT also functions as entropic bristle IDP, the same parameters from SSB were used to characterise NEXT mutants in the library. Mutants with one or more parameters that show less benefit in solubilizing protein compared to original NEXT were discarded. Table 3-4 shows the in-silico selection criteria for NEXT mutants.

Table 3-4. In silico selection criteria of NEXT mutants.

Parameter name	NEXT Criteria
GRAVY	≤ -0.7558
Charge at pH 7	≤ -1.0 or ≥ 1.0
Helix sheet sum	≤ 0.6923
Turn fraction	≥ 0.25
Instability index	≤ 50.35

Overall, 29000 NEXT mutant sequences in library were narrowed to 4285 sequences after in-silico selection.

3.2.2.3 De Novo

As the hydrophilicity of amino acid at each position in a sequence has been predetermined according to the pattern of motifs (see Table 3-2), the GRAVY score and charge at pH 7 was not chosen to characterise De Novo sequences. However, as De Novo sequences were supposed to function as entropic bristle, Helix sheet sum, turn fraction and instability index were still used to characterise and select from De Novo library. Table 3-5 shows the in-silico selection criteria for De Novo sequences.

Table 3-5. In silico selection criteria of De Novo sequences.

Parameter name	De Novo Criteria
Helix sheet sum	≤ 0.5
Turn fraction	≥ 0.4
Instability index	≤ 40.0

Overall, 3747989 De Novo sequences in library were narrowed to 378679 sequences after in-silico selection.

3.2.2.4 CAHS

For CAHS IDP, it was documented that it prevent client protein-protein aggregation by adopting partly an amphiphilic α -helical conformation (especially in motif 1) to associate with and protect aggregation-prone client protein such as lactate dehydrogenase (LDH) (Kang et al., 2024).

Therefore, Pace & Scholtz sum was chosen as lower sum indicates better overall ability of forming such helical conformation to prevent protein-protein aggregation. Similarly, Pace & Scholtz mean was chosen to measure the same ability on a per residue basis (Pace & Scholtz, 1998). Additionally, number of proline residues in each mutant was counted, as more proline residues disrupts the helical conformation more due to its restriction in Φ angle.

Moreover, hydrophobic moment was chosen as higher moment suggests more ability of CAHS mutant to use its hydrophobic side to associate with aggregation-prone patches of hydrophobic residues on surface of client proteins and to use its hydrophilic side to enhance solubility in aqueous cytosol. However, a very high hydrophobic moment may introduce antimicrobial properties into IDP as it causes bacterial membrane disruption (Zhang et al., 2025), and was therefore avoided. Similarly, the fraction of actual hydrophobic residues on the hydrophobic face of the CAHS helix (face occupancy) was chosen as higher face occupancy suggests more CAHS-client protein association that prevents aggregation.

Hence, each CAHS mutant was characterised with these chosen parameters. Then, the mutants were compared against the parameter values from the original CAHS sequence. Mutants with one or more parameters that shows less benefits in solubilizing protein compared to original CAHS were discarded. Table 3-6 lists out the exact in-silico mutant selection criteria for each parameter.

Table 3-6. In silico selection criteria of CAHS mutants.

Parameter	CAHS Criteria
Pace & Scholtz sum	≤ 6.68
Pace & Scholtz mean	≤ 0.352
Hydrophobic moment	$\geq 0.263, \leq 0.526$
Face occupancy	≥ 0.6
Number of proline	≤ 0

Overall, 11000 CAHS mutant sequences in library were narrowed to 458 sequences after in-silico selection.

3.2.3 Compressing selected mutant amino acid sequence into degenerate DNA sequence

Due to limitations on DNA sequence synthesis, we cannot synthesise the exact sequence for each of the selected sequences and mix them into a library. Instead, for each IDP, we compress its selected sequences into a representative DNA sequence with degenerate nucleotides to be synthesised. Degenerate sequence was generated using a customised program and can be accessed in https://github.com/TKA0329/iDEC_Cambridge_2026 .

In short, selected amino acid sequences were first checked for duplicated sequences and deduplicated. Then, the sequences were aligned and a sequence logo was generated to show the type and proportion of different amino acids appearing at each position, as shown in Figure 3-1. The program identifies mutation hotspots, i.e. positions where in-silico selected sequences shows a significant proportion of amino acid different from the original amino acid at that position (the optimal threshold of significant proportion was determined case-by-case for each IDP sequence logo). For positions outside of mutation hotspot, the program keeps only the original amino acid at the position. Within hotspot positions, amino acids showing minimal proportion in the hotspot were excluded. Then, amino acids occupying a large proportion and are different from original amino acid become the target amino acids to be encoded.

Subsequently, all possible 3-letter codons were generated using both degenerate IUPAC nucleotide letters and regular A,T,C,G nucleotide letters. The codon containing degenerate letters (degenerate codon) that encode stop codons were removed. Degenerate codon that represents most target amino acids at the hotspot while minimizing representation of amino acids outside the targets (off-target amino acid) was selected. Various algorithms of optimizing degenerate codon at hotspot were attempted, and direct minimization of target amino acid loss and off-target amino acid gain was chosen. Eventually, degenerate sequence for each IDP was generated as shown by Table 3-7. Figure 3-1 shows the complete schematic workflow of generating degenerate sequence for in-silico selected SSB mutants.

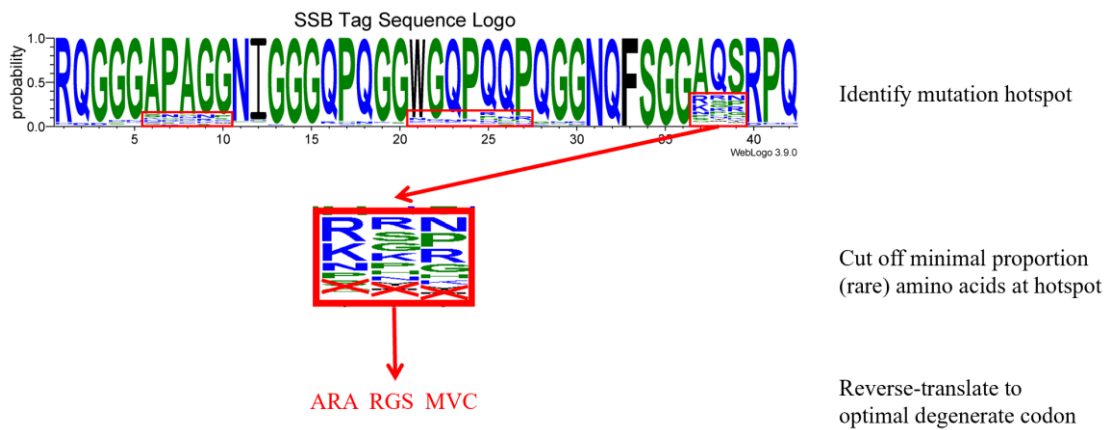


Figure 3-1. Workflow of generating a degenerate DNA sequence encoding the in-silico selected SSB mutants.

3.2.4 Verifying degenerate sequence’s performance

Despite minimizing off-target amino acid gain in designing the degenerate DNA sequence, this degenerate sequence inevitably cannot represent the exact sequence space of the selected mutant sequence. For instance, ACGATA and ATGAGA can be compressed into AYGRTA (Y=C/T, R=A/G). However, AYGRTA also encodes ATGATA and ACGAGA, and all variants encode different amino acids. Hence, it is essential to determine if such by-product sequences also have good in-silico performance in solubility enhancing. Table 3-7 list out the degenerate sequence representing each IDP's library, and table 3-8 list out the estimated number of mutants outperforming the original sequence for each IDP.

Table 3-7. Degenerate sequence of each IDP's library. Symbols of degenerate nucleotides follow IUPAC nucleotide ambiguity code.

[illegible]

Table 3-8. Profile of degenerate sequences in in-silico outperforming original sequence for enhancing protein solubility.

Name	Total number of encoded mutants	Percentage of better mutants	Estimated number of better mutants
SSB	995328	47.3	470332
NEXT	1.41×10^8	97.5	1.38×10^8
De Novo	6.18×10^{50}	4.94	3.05×10^{48}
CAHS	1741824	47.0	819319

From table 3-8, besides De Novo IDP which is fully degenerate and possess larger encoding space, the other 3 degenerate sequences encode nearly 50% or higher percentage of mutants that shows more solubility-enhancing properties based on in-silico prediction. Under comparison, complete random mutagenesis is thought to introduce only 0.01-1% of mutants more beneficial than original variant (Romero & Arnold, 2009). Hence, this computationally aided, semi-random approach of generating mutant library is predicted, under the same proxies, to contain a higher fraction of the desired solubility-enhancing IDP compared to random mutagenesis. Further work to characterise a random library in-silica has to be done to validate this.

Moreover, the number of better mutants for IDPs besides De Novo lies in the magnitude of 10^5 - 10^8 . Based on estimated efficiency of Gibson assembly and E. coli plasmid transformation, this range is desirable as it maximises the diversity of transformed E. coli while minimizing the loss of mutants due to not being Gibson assembled or transformed.

3.3 Experimental generation of complete random library via Error-Prone PCR

While it is suggested that the degenerate sequence libraries of each IDP contains more solubility-enhancing mutants at a higher proportion, relying on limited number of parameters predicted in-silico to determine whether a mutant is solubility-enhancing is partially biased. In order to account for such bias and validate the efficiency of this semi-random library, a completely random library generation by error-prone PCR (epPCR) was carried out in parallel. Given that De Novo IDP does not have an original variant to act as template, epPCR was carried out only on the original sequence template of SSB and NEXT (epPCR on CAHS motif 1 was planned but not carried out due to unavailable template).

The following synthetic dsDNA sequence were used as template, where sequence in [] indicates non-IDP overhang region for primer binding and later for Gibson assembly:

SSB original sequence template:

5'[AGGAAGCCAGACAAATGGTAGT]AGACAAGGAGGAGGAGCGCCGCGGGGGG
GAACATAGGAGGAGGACAACCACAAGGAGGATGGGGGCAGCCACAACAACCAC
AAGGAGGAAACCAATTCAGCGGAGGAGCACAAAGCAGACCACAATAA[AGCTCG
AGCGAAGCTTGGG]3'

NEXT original sequence template:

5'[AGGAAGCCAGACAAATGGTAGT]GCAGTACAACACAGCAACGCACCACTAATA
GACCTAGGAGCAGAAATGAAAAACAACACAAAGAAGCAGCACCAGAAGGAGC
AGCACCAGCACAAAGGAAAAGCACCAGCAGCAGAAGCAAAAAAAGAAGAAGCAC
CAAAACCAAAACCAAGTAGTATAA[AGCTCGAGCGAAGCTTGGG]3'

The following primers were used:

Forward primer (IDP F):

5'AGGAAGCCAGACAAATGGTAGT3'

Reverse primer (IDP R):

5'CCCAAGCTTCGCTCGAGCTTTA3'

Test runs on varying primer concentration, mutagenic Mn^{2+} concentration, Mg^{2+} concentration were first carried out. Subsequently, various annealing temperature, various extension times, and the addition of GC enhancer were tested. The optimised epPCR condition that generates significant amplicon yield while maximizing tolerable mutagenic stress was determined, as shown in Table 3-9 and Table 3-10.

Table 3-9. Composition of optimised 50 μ L epPCR reaction mixture.

Component	Final concentration
Forward primer (IDP F)	0.3 μ mol/L
Reverse primer (IDP R)	0.3 μ mol/L
Template	0.1 mg/L
MnCl ₂ (aq)	0.15 mmol/L
MgCl ₂ (aq)	4 mmol/L
Taq polymerase	2.5 unit / reaction (as defined by ThermoScientific EP0402)
dATP	0.2 mmol/L
dTTP	0.2 mmol/L
dCTP	0.2 mmol/L
dGTP	0.2 mmol/L
10X Taq Buffer with (NH ₄) ₂ SO ₄	1X (as defined by ThermoScientific EP0402)
5X Phusion™ GC Enhancer	1X (as defined by ThermoScientific F631S)

Table 3-10. Thermocycler program for optimised 50 μ L epPCR reaction mixture.

Name	Temperature/°C	Duration/sec	Repeat
Initial denaturation	95	360	1
denaturation	95	60	30
annealing	61.7	30	
extension	72	10	
preservation	4	-	1

30 μ L of product from the first epPCR reaction underwent gel electrophoresis in 2% agarose gel in 1X lithium borate buffer, and was then gel extracted using PureLink™ Quick Gel Extraction Kit (K210012). The template was added to a new epPCR reaction mixture to reach a final concentration of 0.1 mg/L. The epPCR was intended to repeat through such procedure for 8 times to cumulate significant mutations. However, after 3-4 repeats, the mutated template seems unable to be amplified further, and an aliquot was sent for nanopore sequencing to determine variants inside.

Unfortunately, due to time limitation, the Gibson assembly and downstream characterization of epPCR-generated IDP libraries did not take place in this study.

3.4 Rationale in designing selection method.

3.4.1 Rationale in selection method of entropic bristle IDP-- fusing to L76N β -lactamase

To select for an entropic bristle IDP tag's better performance in preventing protein aggregating into inclusion bodies, we proposed various client proteins to fuse with the IDP tag, and their corresponding method to measure solubility. One method was fusing IDP to aggregation-prone variety of GFP protein and measure increment in fluorescence intensity due to more solubilised GFP. Another method was fusing IDP to aggregation-prone interleukin-3 (IL3) and apply densitometric measurement on bands of IL3 in soluble fraction and in inclusion bodies on a SDS-PAGE gel. However, these 2 methods are low in throughput as the fluorescence or optical density must be manually measured for each individual mutant. It would be time-consuming to manually measure and pick out the colony with best IDP. Therefore, to automate the selection with high throughput, we linked the solubility-enhancing property of IDP tag to the survival of *E. coli*.

Eventually, we decided to fuse entropic bristle IDPs to L76N mutant of TEM-1 β -lactamase and subject the expressing *E. coli* to β -lactam antibiotic. Beta-lactamase is a type of enzyme that catalyzes the hydrolytic breakdown of β -lactam antibiotics (such as ampicillin or carbenicillin) and confers antibiotic resistance to bacteria. The L76N variant of TEM-1 β -lactamase is aggregation-prone and forms periplasmic inclusion bodies significantly more than wild type β -lactamase (Sideraki et al., 2001). Sideraki et al. (2001) also showed that additional mutations that make aggregation of L76N mutant kinetically unfavored can rescue the *E. coli*'s resistance to β -lactam antibiotics. An entropic bristle IDP tag can also kinetically disfavor protein aggregation kinetically, because aggregation-prone intermediate cannot quickly aggregate into inclusion bodies, with the IDP tag excluding other β -lactamase in the intermediate's surrounding.

Hence, under appropriate β -lactam antibiotic concentration, if a good solubility-enhancing IDP tag is fused to β -lactamase, more β -lactamase end up in functional, soluble form. This catalyzes more breakdown of antibiotic to provide the *E. coli* with a survival advantage in the presence of β -lactam antibiotic. On the contrary, if an IDP tag is poor at solubility-enhancing, more β -lactamase readily end up more as non-functional form in inclusion bodies. This weakens the *E. coli*'s resistance to β -lactam antibiotic, and the *E. coli* would die of osmotic lysis due to inhibited cell wall synthesis. Figure 3-2 shows an outline of how solubility-enhancing property of IDP tag was linked to bacterial survival rate.

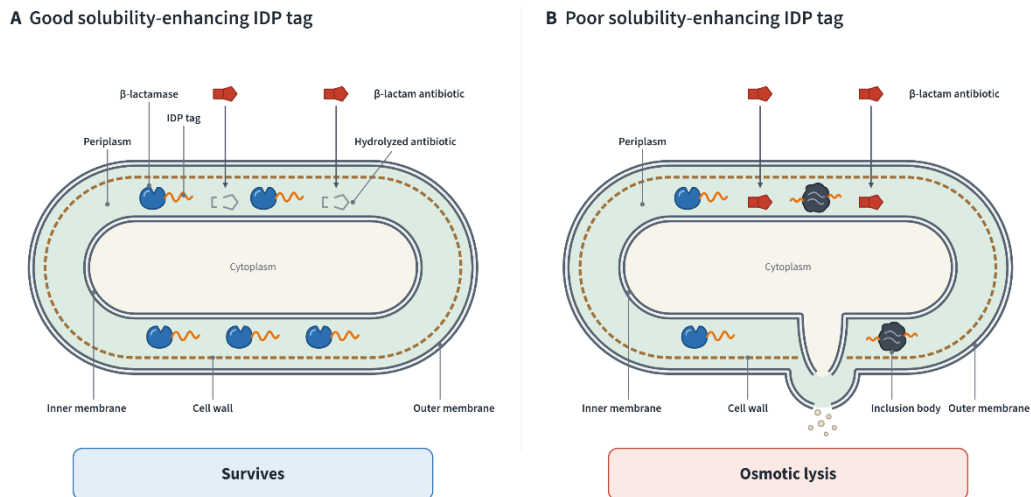


Figure 3-2. Schematic diagram showing entropic bristle IDP selection via fusing IDP to L76N TEM-1 β -lactamase and subjecting the expressing *E. coli* to β -lactam antibiotic stress. Diagram generated from Claude AI and manually verified.

3.4.2 Rationale in selection method of CAHS IDP--freeze-thaw cycles

We decided to express CAHS IDPs as individual protein instead of fused tags and subject the expressing *E. coli* to repeated freeze-thaw cycles. Kang et al. (2024) reported that repeated freeze-thaw cycles lead to more aggregation of proteins, such as lactate dehydrogenase. Moreover, CAHS protein does not function by being covalently linked to client protein as, and it associates non-specifically to aggregation-prone proteins to shield them from aggregating.

Therefore, we reasoned that CAHS IDPs expressed in *E. coli* can non-specifically associate with and protect various cytoplasmic proteins that are both aggregation-prone during repeated freeze-thaw and critical for *E. coli* survival. This selection method is also high-throughput as it links the solubility-enhancing performance of CAHS IDP mutants to the survival of *E. coli*, and no individual measurement of solubility for each CAHS IDP mutant is needed.

3.5 Cloning

3.5.1 Entropic-bristle IDP cloning

As β -lactamase N-terminus contains signaling peptide sequence for it to be transported correctly into periplasm, the entropic bristle IDPs can only be fused to its C-terminus to avoid changing β -lactamase secretion and decreasing β -lactam antibiotic resistance.

Besides, in expression of β -lactamase-IDP fusion protein, the arabinose-inducible promoter from araBAD operon was chosen. Under induction of 0.2% L-arabinose, the expression level of protein under arabinose-inducible promoter reaches approximately 20% of total cell protein (BioProcess Tools Team, 2026). This expression level of β -lactamase is higher than that under the natural promoter of β -lactamase used by Sideraki et al. (2001). Therefore, we expected this higher level of β -lactamase to cause more aggregation into inclusion body, which imposes greater selection pressure for the fused entropic bristle IDP tags to prevent aggregation. Moreover, the arabinose-inducible promoter does not confer expression level as high as T7 promoter, so not all β -lactamase aggregates into inclusion bodies.

With these considerations, the plasmids were constructed as shown in Figure 3-3. First, the original plasmid contains an sfGFP ORF under the control of arabinose-inducible promoter, which needs to be removed. This was achieved by PCR amplifying the plasmid with the forward primer “5-AGCTCGAGCGAAGCTTGGGCCCCGAACAA-3” and reverse primer “5-GGTAAATTCCTCCTGTTAGCCCCAAAAACG-3”. The amplicon is a linearized pBAD plasmid with sfGFP ORF removed (linear-pBAD-no-sfGFP). The amplicon was then digested with DpnI (Thermo Fisher Scientific ER1701) to remove the circular plasmid templates. The amplicon was then purified using GenElute™ PCR Clean-Up Kit (Sigma-Aldrich, NA1020).

Then, the dsDNA encoding L76N TEM-1 β -lactamase and a C-terminal GGGSGGGGS linker was purchased from Integrated DNA Technologies: “5-[GGGCTAACAGGAGGAATTAACC]ATGAGTATTCAACATTTCCGTGTCGCCCTTATTCCTTTTTTGCGGCATTTCCTTCCTGTTTTTGCTCACCCAGAAACGCTGGTGAAAGTAAAAGATGCTGAAGATCAGTTGGGTGCACGAGTGGGTACATCGAACTGGATCTCAACAGCGGTAAGATCCTTGAGAGTTTTCGCCCCGAAGAACGTTTTCCAATGATGAGCACTTTTAAAGTTCTGAACTGTGGCGCGGTATTATCCCGTGTTGACGCCGGGCAAGAGCAACTCGGTCGCCGCATACACTATTCTCAGAATGACTTGGTGAGTACTCACAGTCACAGAAAAGCATCTTACGGATGGCATGACAGTAAGAGAATTATGCAGTGTGCCATAACCATGAGTGATAACACTGCGGCCAACTTACTTCTGACAACGATCGGAGGACCGAAGGAGCTAACCGCTTTTTTGCACAACATGGGGGATCATGTAACCTGCCTTGATCGTTGGGAACCGGAGCTGAATGAAGCCATACCAAACGACGAGCGTGACACCACGATGCCTGCAGCAATGGCAACAACGTTGCGCAAACCTATTAACCTGGCGAACTAC

TTACTCTAGCTTCCCGGCAACAATTAATAGACTGGATGGAGGCGGATAAAGTTGCA
GGACCACTTCTGCGCTCGGCCCTTCCGGCTGGCTGGTTTATTGCTGATAAATCTGG
AGCCGGTGAGCGTGGGTCTCGCGGTATCATTGCAGCACTGGGGCCAGATGGTAAG
CCCTCCCGTATCGTAGTTATCTACACGACGGGGAGTCAGGCAACTATGGATGAACG
AAATAGACAGATCGCTGAGATAGGTGCCTCACTGATTAAGCATTGGGGAGGAGGA
GGAAGCGGAGGAGGAGGAAGCCAGACAAATGGTAGTTAA[AGCTCGAGCGAAGC
TTGGGCCCCG]-3” This dsDNA contains Gibson assembly homology arms as indicated in [],
so this dsDNA was inserted into linear-pBAD-no-sfGFP using Gibson Assembly[®] Master Mix
(E2611S) from New England Biolabs. The molar ratio between linear-pBAD-no-sfGFP and the
insert was 1:2, and the incubation was 15 minutes at 50 °C. The product plasmid (pBAD-AmpR,
see middle panel of Figure 3-3) expresses L76N TEM-1 β -lactamase without entropic bristle
IDP tags fused to C terminus, and acts as a control to compare against β -lactamase fused with
IDP.

Furthermore, pBAD-AmpR underwent PCR amplification using forward primer “5-
TAAAGCTCGAGCGAAGCTTGGGCCCCGAACAAAACTCATC-3” and reverse primer “5-
ACTACCATTGTCTGGCTTCCTCCTCCTCCGCTTCCTCCTCCTCCCCAATGCTTAATC
AGTG-3”. The amplicon is a linearized pBAD-AmpR with no sequence excluded, and the
linearization break point is immediately before the STOP codon of the inserted β -lactamase +
GGGSGGGGS linker open reading frame (linear-pBAD-AmpR). This amplicon was digested
again with DpnI and PCR-purified using the same procedure. Then, all original and degenerate
sequences of entropic bristle IDPs (as listed below) were purchased from IDT:

Original SSB sequence:

5-
[AGGAAGCCAGACAAATGGTAGT]AGACAAGGAGGAGGAGCGCCGGCGGGGGG
AACATAGGAGGAGGACAACCACAAGGAGGATGGGGGCAGCCACAACAACCACA
AGGAGGAAACCAATTCAGCGGAGGAGCACAAAGCAGACCACAA[TAAAGCTCGA
GCGAAGCTTGGG]-3

Original NEXT sequence:

5-
[AGGAAGCCAGACAAATGGTAGT]GCAGTACAACACAGCAACGCACCACTAATAGA
CCTAGGAGCAGAAATGAAAAACAACACAAAGAAGCAGCACCAGAAGGAGCAG
CACCAGCACAAAGGAAAAGCACCAGCAGCAGAAGCAAAAAAAGAAGAAGCACCA
AAACCAAAACCAGTAGTA[TAAAGCTCGAGCGAAGCTTGGG]-3

Degenerate SSB sequence:

5-

[AGGAAGCCAGACAAATGGTAGT]AGACAAGGAGGAGGAARCCSAGVCRGARGCA
ACATAGGAGGAGGACAACCACAAGGAGGAAASCRACRACVCCVAMASCSACAAG
GAGGAAACCAATTCAGCGGAGGAARARGSMVCAGACCACAA[TAAAGCTCGAGC
GAAGCTTGGG]-3

Degenerate NEXT sequence:

5-

[AGGAAGCCAGACAAATGGTAGT]GCAGTACAACACAGCAACGSAARCCYAADCG
ACCTAGGAGCAGAAATGAAAAACAACACAAVMSMRAAASCCAGAAGGAVVC
GSAARCGCACAAGGAAAAGCARRSRMSHACRASRVAAAAAAGAAGAAGCACCA
AAACCAAAACCAGTAGTA[TAAAGCTCGAGCGAAGCTTGGG]-3

Degenerate De Novo sequence:

5-

[AGGAAGCCAGACAAATGGTAGT]VRSVRSVASVASVRSVRSVRSRRSVASVRSVNW
VDWVRSVRSVRSVRSVRSVSVSVASVRSVNWNWVRSVRSVRSVRSVSVSVRSV
RSVNWNWVRSVRSVASVRSVRSVSVSVRSVRSVNWNWVSVRSVASVASVRSVR
SVRSRRSVNWNW[TAAAGCTCGAGCGAAGCTTGGG]-3

All of these sequences contain Gibson assembly homology arms, so each of them was inserted into linear-pBAD-AmpR using Gibson Assembly[®] Master Mix (E2611S) from New England Biolabs. The molar ratio between linear-pBAD-AmpR and the insert was 1:5, and the incubation was 30 minutes at 50 °C. The product plasmid (pBAD-AmpR-IDP, see right panel of Figure 3-3) expresses the L76N TEM-1 β -lactamase with original or degenerate IDP sequences fused to the C-terminus after the GGGGSGGGGS linker sequence.

One additional step was done to the three degenerate IDP sequences before Gibson assembly. The degenerate sequences can only be synthesized as single strand DNA, so a customized protocol (See table 3-11, table 3-12) was applied to convert ssDNA into dsDNA before assembly.

Table 3-11. Composition of reaction mixture for converting ssDNA degenerate IDP sequences to dsDNA.

Component	Final concentration
Forward primer (IDP F, same as epPCR)	0.5 μ mol/L
Reverse primer (IDP R, same as epPCR)	0.5 μ mol/L
ssDNA degenerate IDP template	0.1 mg/L
Q5 [®] -XT Hot Start High-Fidelity 2X Master Mix (M2499S)	1X (as defined by NEB M2499S)

Table 3-12. Thermocycler program for 50 μ L ssDNA to dsDNA conversion reaction mixture.

Name	Temperature/°C	Duration/sec	Repeat
Initial denaturation	98	30	1
denaturation	98	30	30
Annealing	61.7	20	
Extension	72	7	
preservation	4	-	1

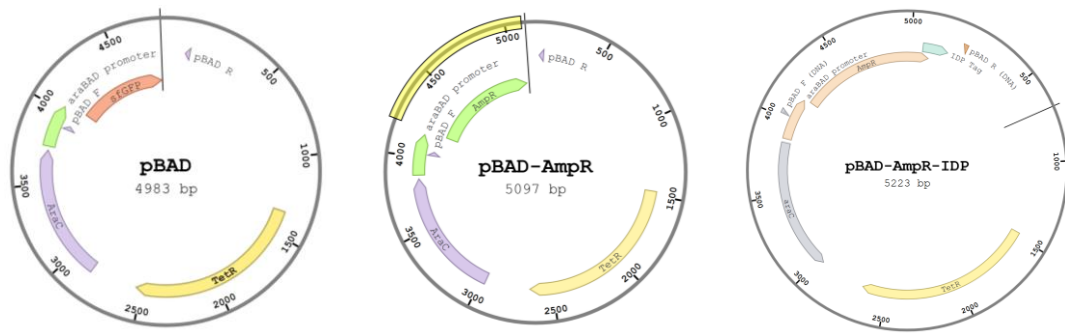


Figure 3-3. Maps of annotated plasmids transformed into E coli for selection on entropic bristle IDP. The left panel (pBAD) shows the original plasmid purchased. The middle panel (pBAD-AmpR) shows plasmid expressing L76N TEM-1 β -lactamase without entropic bristle IDP fused to it. The right panel (pBAD-AmpR-IDP) shows plasmid expressing L76N TEM-1 β -lactamase with entropic bristle IDP fused to C terminus.

3.5.2 CAHS IDP cloning

The original CAHS IDP protein gene was purchased as dsDNA from IDT with the following sequence:

“5-[GGGCTAACAGGAGGAATTAACC]ATGACTGAAGCTTATCGCAAACAGCAGGAAGT GGAAGCCGACAAGATCCGCAAAGAGCTGGAGAAGCAGCACCTGCGCGATGTGGA GTTCCGCAAAGACATCGTCGAAATGGCCATTGAAAACCAGAAAAAAATGATTGAC GTGGAGAGCCGCTATGCCAAAAAAGATATGGACCGCGAGCGTGTGAAAGTGTAAGCTCGAGCGAAGCTTGGGCCCCG]-3”. The sequence contains Gibson assembly homology arm indicated in []. Therefore, the sequence was inserted into linear-pBAD-no-sfGFP using Gibson Assembly[®] Master Mix (E2611S) from New England Biolabs. The molar ratio between linear-pBAD-no-sfGFP and the insert was 1:5, and the incubation was 15 minutes at 50 °C. The product plasmid (see Figure 3-4) expresses the CAHS IDP protein alone inside cytoplasm.

The degenerate CAHS IDP was too long to be synthesized in 1 DNA fragment, so it was split into fragment motif 1 carrying degenerate nucleotides and fragment 2 with constant nucleotides only.

Fragment motif 1: “5-[GGGCTAACAGGAGGAATT]AACCATGACAGAAGCAAKRMKGARAMRAATGRVAA

RAGMAADGRRSARARTAAGAAAASAAWTGRMAAAAATG[CACCTAAGAGACGTA
GAATTCAG]-3”, fragment motif 2: “5-
[CACCTAAGAGACGTAGAATTCAG]AAAAGACATAGTAGAAATGGCAATAGAAAAC
CAAAAAAAAAATGATAGACGTAGAAAGCAGATACGCAAAAAAAGACATGGACAGA
GAAAGAGTAAAAGTATAA[AGCTCGAGCGAAGCTTGGGCCCCG]-3”. Gibson assembly
homology arms are indicated in [].

As fragment motif 1 was also synthesized in single strand, the same ssDNA to dsDNA conversion protocol outlined in Table 3-11 and Table 3-12 was applied using forward primer “5-GGGCTAACAGGAGGAATTAACC-3” and reverse primer “5-CTGAATTCTACGTCTCTTAGGTGC-3”. Subsequently, a 3-piece Gibson assembly between linear-pBAD-no-sfGFP, fragment motif 1, and fragment 2 was carried out using NEBuilder® HiFi DNA Assembly Master Mix (E2621S). The molar ratio of the 3 components was 1:5:5. Linear-pBAD-no-sfGFP and fragment 2 were first incubated together for 30 minutes at 50 °C, then fragment motif 1 was added and incubated further for 30 minutes. The product plasmid is the same as Figure 3-4, where CAHS1 becomes the degenerate CAHS IDP. Alternative method of Gibson assembly was later developed using NEBuilder® HiFi DNA Assembly Master Mix (E2621S), where dsDNA linear-pBAD-no-sfGFP, ssDNA fragment motif 1 and dsDNA fragment 2 were added together in molar ratio 1:200:5 and incubated for 60 minutes at 50 °C. Product plasmid was still the same as in Figure 3-4.

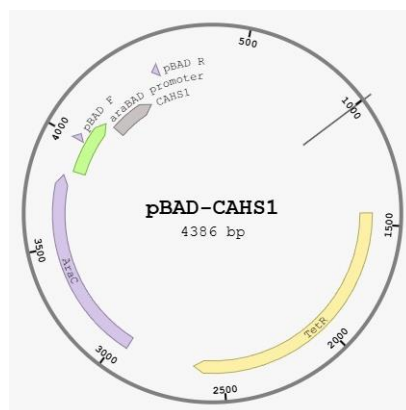


Figure 3-4. Map of annotated plasmid transformed into E coli for selection on CAHS IDP.

3.6 Transformation

pBAD-AmpR-IDP and pBAD-CAHS1 libraries were transformed into competent DH10β E. coli cells. For each library, 50ul frozen glycerol stocks of competent DH10β E. coli cells were thawed on ice, and 2μl of pBAD-IDP Gibson assembly products were added. The cells were rested on ice for 30 minutes, then subjected to heat shock at 42°C for 30 seconds. The cells underwent recovery in SOC media for 1 hour before being incubated overnight in LB media culture.

3.7 Protein expression induction

The expression of β -lactamase-IDP fused proteins was induced through the addition of 0.2% L-arabinose. Expression of CAHS IDP was induced with 1% L-arabinose.

3.8 Beta-lactam antibiotic selection

All OD600 readings were taken in a 96-well plate reader with 100 μ L triplicates. All data reported is not normalised to the standard 1 cm cuvette pathlength but can be converted with a multiplication factor of 5.6.

Induced cell cultures containing original and derived mutant libraries of SSB, NEXT and De Novo IDP tags were added to Lennox LB media containing 0.2% L-arabinose and 10 μ g/mL tetracycline to make up to varying concentrations of carbenicillin, along with cell cultures containing empty pBAD vectors and pBAD containing β -lactamase inserts. The final concentrations of carbenicillin in cell cultures were 0, 5000, 10000, 15000, 30000 and 40000 μ g/ml, as determined by previous experiments investigating the rate of cell survival in varying carbenicillin concentrations. The cell cultures were made up to OD600 0.05 to normalise starting OD. They were incubated at 37°C with shaking, and OD600 readings were taken at regular intervals as proxies for population count.

Surviving cells after 21 hours of incubation with 0 and 40000 μ g/ml carbenicillin were plasmid extracted.

3.9 Repeated freeze-thaw selection

3.9.1 Flash freeze selection workflow

Induced cell culture suspensions containing the empty pBAD vector, CAHS derived library sequences and the wild type CAHS sequence were divided into 1.0 mL aliquots and subjected to 6 consecutive rounds of flash freeze selection, along with tubes that were not subjected to freeze-thaw selection (T0). For each cycle, each cell culture suspension was immersed in isopropanol/dry ice bath for 15 minutes to ensure complete freezing before being thawed at room temperature for 25 minutes. Cell culture suspensions were inspected visually between cycles to ensure complete thawing before the next stage of freezing and plated immediately after serial dilutions.

3.9.2 Deep freeze selection workflow

An additional freeze-thaw assay was performed, with cells frozen in a -70 °C freezer rather than the dry-ice/isopropanol cryogenic bath used in the flash-freeze assay.

Induced cell culture suspensions containing the empty pBAD vector, CAHS library sequences and the wild type CAHS sequences were divided into 0.5mL aliquots and subjected to deep freeze at -70°C . Samples were frozen overnight with no strict timing due to no expected difference in mortality based on cumulative time spent frozen. (Sleight, Wigginton and Lenski, 2006) The cells were thawed for 45 minutes, serially diluted then plated. Colony count was used as a proxy for cell survival after selection.

3.9.3 Excluding contaminating pBAD colonies

To distinguish putative library-containing colonies from an empty-vector background, cell suspensions were plated on agar supplemented with tetracycline and L-arabinose. The empty pBAD vector retained an arabinose-inducible sfGFP reporter, enabling colonies carrying this vector to be identified by fluorescence under UV illumination. Fluorescent colonies were excluded from colony counts used to assess the survival of the CAHS library. Non-fluorescent colonies were counted as putative insert-containing colonies; insert identity was not confirmed by colony PCR.

3.10 Sequencing on selected variant

Cell cultures that had undergone carbenicillin were incubated at 25°C whilst freeze-thaw selection samples were incubated at 37°C . Both cell cultures were incubated until they reached below saturation based on OD600 reading data. A lower incubation temperature was chosen to minimise the risk of oversaturation providing additional selection pressure that may skew the variant representation in the libraries. Plasmids were isolated from each cell culture using Thermofisher Purelink Quick Plasmid Miniprep Kit (K210010).

We hope to obtain as many sequencing reads as possible for greater coverage of variants after selection. Therefore, we chose to use linear amplicon sequencing instead of whole plasmid sequencing. Hence, a customized PCR protocol was developed to amplify regions containing IDP region without changing the proportion of different variants. Forward primer used (pBAD F, see Figure 3-3): “5- GTGTCTATAATCACGGCAGAAAAGTCCACA-3”. Reverse primer used (pBAD R, see Figure 3-3):” 5-TCATCCGCCAAAACAGCCAAGCTGGAGACCG-3”. This was achieved using a longer period of complete denaturation in each cycle and reducing the number of amplification cycles. Table 3-13 and Table 3-14 show the composition and condition of such PCR.

Table 3-13. Composition of customized PCR to extract IDP-containing linear amplicon.

Component	Final concentration
Forward primer (pBAD F)	0.3 $\mu\text{mol/L}$
Reverse primer (pBAD R)	0.3 $\mu\text{mol/L}$

Plasmid extracted after selection	1.0 mg/L
Q5 [®] -XT Hot Start High-Fidelity 2X Master Mix (M2499S)	1X (as defined by NEB (M2499S))

Table 3-14. Condition of customized PCR to extract IDP-containing linear amplicon.

Name	Temperature/°C	Duration/sec	Repeat
Initial denaturation	98	60	1
Denaturation	98	60	20
Annealing	65	20	
Extension	72	16	
Preservation	4	-	1

Gel electrophoresis was conducted on the amplicon using 2% agarose gel in 1X lithium borate solution at 120V for 90 minutes. The correct amplicon was gel extracted using PureLink™ Quick Gel Extraction Kit (K210012). Amplicons were sequenced by Oxford Nanopore R10.4.1.

3.11 Sequencing data processing

Raw reads obtained from Plasmidsaurus Amplicon sequencing in the form of a .fastq file were analysed to extract variant identity and distribution within the population.

Raw amplicon sequencing reads were provided in FASTQ format, containing DNA sequences and their corresponding Phred quality scores. A custom Python-based analysis pipeline was developed to process the reads, identify the variable desired region, perform sequence-level quality control, rescue selected reads containing indels, and quantify the resulting DNA and amino acid variants.

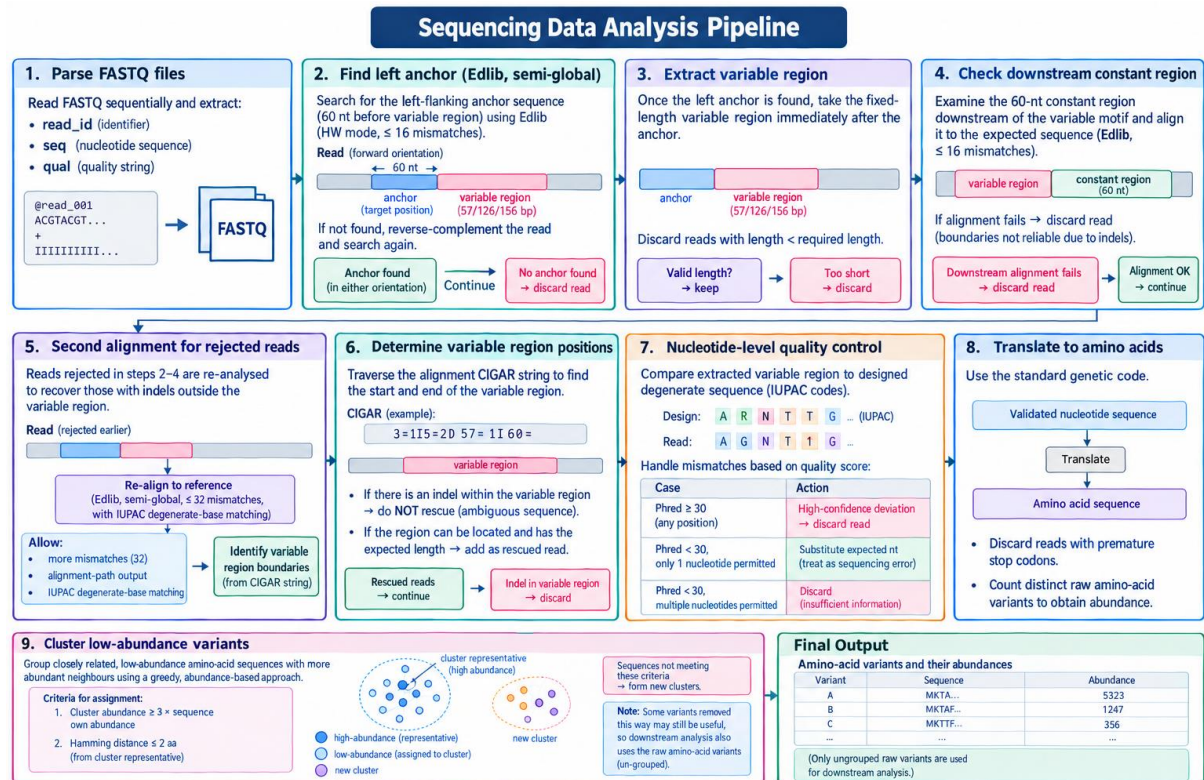


Figure 3-5. Diagram summary of sequence data analysis pipeline (Generated by Chat-GPT)

The order the pipeline adopts is as follows:

1. FASTQ files were parsed sequentially to extract the read identifier ('read_id'), nucleotide sequence ('seq') and quality string ('qual').
2. To identify the region of interest independently of sequencing orientation, each read was searched for a defined left-flanking anchor sequence 60 nucleotides ahead of where the desired region is projected to start using the Edlib sequence-alignment library. A semi-global alignment (HW) was used so that the anchor can be identified within a longer sequencing read without requiring alignment of the entire read. Up to 16 differences were permitted during this initial search. If the anchor was not identified in the original orientation, the read sequence and quality string were reverse-complemented, and the search was repeated. Reads for which the left anchor could not be identified in either orientation was excluded from subsequent analysis.
3. Once the left anchor has been identified, the variable region is defined as the fixed-length nucleotide sequence (57 bp for PrCAHS1-1, 126 bp for SSB, and 156 bp for NEXT) immediately following the end of the anchor. Reads that have lengths smaller than the fixed length are discarded.
4. To identify reads in which an insertion or deletion could have corrupted the expected boundaries on the right, the 60-nucleotide constant region immediately downstream of the variable motif was also examined. This region was aligned against the expected

downstream sequence using Edlib with a tolerance of up to 16 mismatches. Reads for which this downstream sequence could not be aligned within the specified tolerance were excluded, as the expected motif boundaries could no longer be considered reliable for extracting the desired region.

5. Reads rejected during the initial anchor-based processing were subsequently subjected to a second alignment procedure to identify reads that could potentially be recovered if the insertions and deletions occur outside of the desired region.
6. Each candidate read was aligned to this reference using a semi-global alignment with Edlib's alignment-path output. The alignment was permitted by a larger error tolerance (32 mismatches) to accommodate reads containing indels or other sequencing errors. IUPAC degenerate-base equivalences were incorporated into the alignment so that positions represented by degenerate bases in the designed sequence could match any of their permitted nucleotide identities.
7. The alignment CIGAR string was then traversed to determine the positions corresponding to the beginning and end of the variable region in the read. Reads containing an indel within the variable region itself were not rescued, because the resulting nucleotide sequence could not be assigned unambiguously to the predefined region. Reads in which the region could be located and retained at its expected length were added to the dataset as rescued reads.
8. The extracted variable regions were subsequently compared against the designed degenerate nucleotide sequence. IUPAC nucleotide codes were used to define the permitted nucleotide identities at each position. For example, positions represented by a non-degenerate base were required to contain that specific nucleotide, whereas degenerate positions were permitted to contain any nucleotide represented by the corresponding IUPAC code.
9. A mismatch at a position was treated differently according to the associated base quality. Mismatches with a Phred quality score of ≥ 30 were considered high-confidence deviations from the designed sequence and the read was discarded. For a mismatch with a Phred score below 30 at a position where only one nucleotide was permitted, the expected nucleotide was substituted to the correct one, treating the observation as a likely sequencing error. In contrast, low-quality mismatches at positions permitting multiple nucleotide identities were discarded because the available information was insufficient to determine which permitted nucleotide had been correctly sequenced.
10. Reads passing the nucleotide-level quality control criteria were translated into amino acid sequences using the standard genetic code. Reads producing a premature stop of codon were excluded from the final amino-acid sequences, and the remaining reads were counted to obtain the abundance of each distinct raw amino-acid variant.
11. To reduce the influence of residual sequencing errors, closely related low-abundance amino-acid sequences were grouped with substantially more abundant neighbouring variants using a greedy, abundance-based clustering procedure. A lower-abundance

sequence was assigned to an existing cluster when the cumulative abundance of that cluster was at least three times greater than its own abundance and its hamming distance from the cluster representative was no greater than two amino acids substitutions. Sequences not satisfying these criteria formed new clusters. Nevertheless, we acknowledged that some of the variants discarded this way can still be useful, so the subsequent analysis of the sequencing result uses raw amino acid variants that have not undergone this grouping.

3.12 Analysis of change in population composition

Amino acid variant counts obtained from sequencing were analysed separately for each tag derived library. Within each library, variants were matched by their complete amino acid sequence, and the union of sequences from before and after selection were used to construct a two-row contingency table. Columns represented individual variants with 0 assigned where a variant was only detected in a single sample. All variants retained in the extracted sequencing data were preserved without restricting analysis to the most abundant sequences.

Differences in composition were quantified using chi-squared test under the null hypothesis that the two populations before and after selection had equal variant compositions:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where O is the observed variant frequency and E is the expected variant frequency under the null hypothesis.

As most variants were observed to have low expected counts, statistical significance was assessed using a Monte Carlo simulation. For comparison, 100,000 random contingency tables were generated using Patefield's algorithm, preserving the total number of sequences overall and combined counts of each variant. Chi-squared values were calculated for each simulated table, and the Monte Carlo p-value was estimated as follows:

$$\frac{b + 1}{B + 1}$$

where b is the number of simulated tables with chi-squared values at least as the observed statistics, and B is the number of simulations. For reproducibility, a random seed of 42 was used.

The exact binomial confidence intervals were calculated to quantify the level of uncertainty arising from a finite amount of simulations, and p-value was reported at simulation resolution limit when no simulated table had a greater chi-squared value than the observed data.

As single biological replicates, these tests characterise changes in composition between samples but do not establish the reproducibility of enrichment or distinguish the effects of selection from other factors such as founders' effect.

Population mean and standard deviations for each specific property were calculated and compared through Mann-Whitney U test to determine statistical significance of the difference between pre and post treatment data.

Top performing tags from each library were chosen based on the percentage increase of representation within the population after selection. For CAHS1-derived tags, the sum of percentage changes between rounds 1-3 and 3-6 were considered.

3.13 Clustering of amino acid variants

Hierarchical clustering was used to investigate any changes in representation of variants with similar properties before and after selection. A shared pool of amino acid sequences were constructed from the untreated and selected samples within each library. Each sequence was represented once for cluster fitting while its read counts in both samples were retained separately. This approach provided consistent cluster definitions across conditions and prevented common sequences from dominating the definition of property clusters.

Sequences were characterised by the same descriptors used to evaluate variants during sequence generation. For SSB and NEXT derived libraries, this consists of GRAVY score, estimated net charge at pH 7, instability index, turn fraction and the sum of helix and sheet fractions. IDP-BERT, a transformer model for predicting IDP properties (Mollaei et al., 2024), was used to characterise the predicted radius of gyration, heat capacity and end-to-end decorrelation time. For the CAHS1 derived library, Pace-Scholtz sums and means, hydrophobic moments, helical face occupancy, number of salt bridges, prolines and glycines were characterised. Features within each sample were standardised to 0 mean and unit variance to prevent the differences in numerical scale dominating the calculation of Euclidean distances between points.

Agglomerative hierarchical clustering was performed with Ward linkage and Euclidean distance. Partitions containing 2 to 9 clusters were generated and evaluated using mean silhouette scores and the highest scoring partition was selected. The formula for calculating silhouette score is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where $a(i)$ is the average distance of the point i from every point in its cluster, and $b(i)$ is the average distance of the point i from every point in the nearest cluster.

Cluster selections were interpreted as descriptions of sequence properties rather than established functional classes within each sample.

Dimension reduction was performed through principal component analysis for visualisation. Sequences were plotted on the first two principal components with shaded areas proportional to relative sequence abundance. Read weighted cluster centres were displayed to illustrate changes in the composition and properties of member sequences.

3.14 Validation

The validation rounds involve expressing tags with desired target proteins in DH10 β E. coli cells and purifying the protein products to carry out solubility tests and enzyme activity assays.

3.14.1 Validation of CAHS1 sequences

Variants recovered from freeze-thaw cycles 0, 3 and 6 were sequenced with Plasmidsaurus Amplicon Sequencing (Oxford Nanopore, R10.4.1) to identify surviving sequences before and after the exertion of selection pressure.

Surviving variants were evaluated computationally using CALVADOS, a coarse-grained implicit-solvent model implemented with OpenMM. For the wild-type, a shuffled-sequence control and the surviving variants, we assessed their contacts per chain, the largest cluster fraction and the mean radius of gyration of the variants in various conditions. This is done by performing statistical analysis on the trajectories generated from the simulation.

Further work to validate the performance of variants can be done via LDH assays. Prior work from Kang et al. (2024), demonstrated that the CAHS Core – the same region targeted in this study – protects LDH activity during both dehydration-rehydration and freeze-thaw treatments, which means that a similar experiment could be performed on the selected variants to compare their effectiveness against the CAHS Core.

3.14.2 Solubility assay of commercially available solubility tags

Two commercially available protein solubility tags were investigated for their ability in solubilising recombinant proteins during expression to form a baseline for establishing the performance of evolved IDP tags. The Maltose Binding Protein (MBP) tag is a ~45kDA monomeric protein fusion tag partner that increases passenger protein solubility through chaperone-like interactions that prevent intermolecular aggregation. A smaller protein fusion

partner, the B1 domain of *Streptococcal* protein G (GB1) was also identified for its small size of 56 amino acids, comparable to that of evolved IDP tags.

Three aggregation prone proteins were identified to be fused with MBP and GB1 to evaluate protein solubility. *Thermovibrio ammonificans* derived carbonic anhydrase (taCA) was identified for its commercial utility, tendency for aggregation in *E. coli* expression and prior work in increasing solubility with the NEXT tag (Jo, 2024).

Ideonella sakaiensis derived PETase (isPETase) and recombinant human epidermal growth factor (hEGF) were similarly identified for aggregation prone before during *E. coli* expression.

Gene fragments containing each protein attached to each tag with a histidine-6 tag at the C terminus were obtained through IDT gBlocks Gene Fragments service. All gene fragments were cloned into pBAD vectors using NEB Hifi DNA Assembly kits according to manufacturer instructions.

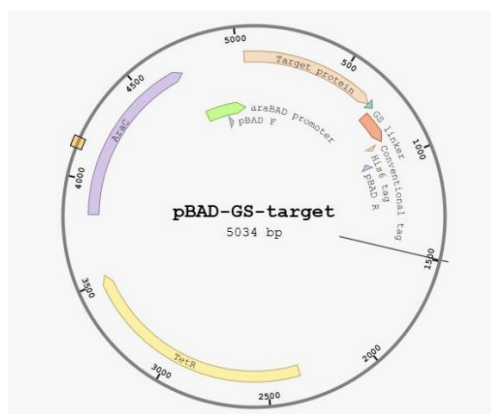


Figure 3-6. Generalized map of annotated plasmid expressing more aggregation-prone target proteins with IDP and His6 fused to C terminus.

3.14.3 Protein synthesis and extraction

pBAD vectors containing gene inserts were transformed into chemically competent BL21 *E. coli* cells. Cell cultures were incubated overnight, then induced with 1% L-arabinose and further incubated for 6 hours. The cell pellet was suspended in a centrifuge for 20 minutes at x4k g.

E. coli cells were lysed through chemical lysis by resuspension of cell pellet in 1 mL of lysis buffer (20 mM Na₃PO₄, 500 mM NaCl, 20mM imidazole, 0.2 mg/mL Lysozyme, 20 µg/mL DNase A, 1 mM MgCl₂, 1mM PMSF). The lysate is then lysed via mechanical lysis through three freeze-thaw cycles and three bead disruption cycles to obtain the cell lysate with minimal disruption of the inclusion bodies.

Protein-tag complexes were isolated and purified using Cytiva His SpinTrap column. Wash buffer 1 (10 mM Na_2HPO_4 , 10 mM $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, 500 mM NaCl), wash buffer 2 (10 mM Na_2HPO_4 , 10 mM $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, 500 mM NaCl, 20 mM imidazole) and elution buffer (10 mM Na_2HPO_4 , 10 mM $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, 500 mM NaCl, 500 mM imidazole) was made up.

Cytiva His SpinTrap column was prepared according to manufacturer instructions using wash buffer 1. 500 μL of each sample was spun down, the insoluble fraction was unable to be analyzed. 50 μL of supernatant was reserved whilst the other parts were purified through the spin column. Three washes each of wash buffer 1 & 2 were conducted with the flow-through reserved, whilst two washes of the elution buffer were conducted and flow-through reserved.

5 μL of the supernatant, washings and elutions were loaded after 30 minute incubation on a 12% SDS-Page Tris-Tricine gel, along with 1 μL of 500 mM DTT, 2.5 μL of LDS Sample Buffer, and 1.5 μL of H_2O , and the elutions were also loaded to compare via densitometric analysis. Samples were ran at 90V for 100 minutes, fixed in 50% methanol and 10% acetic acid for 20 minutes, stained in Instant Blue overnight and destained in water for an hour before imaged.

3.14.4 Densitometric Analysis

The images were analyzed on ImageJ using the measurement tool and normalised to identified housekeeping band. The pixel brightness is measured and the signal is obtained via $(\text{BG mean} - \text{Band mean}) \times \text{Area}$. Values were normalised by dividing protein band signal by housekeeping band signal. Values are treated as proxies for protein amounts to be compared to judge relative improvements in solubility.

3.14.5 In-Silico Validation of Tags

The parameters used by our in-silico selection of the mutant variants were validated as good proxies by comparing the bounds used in selection against the average parameter score of our post-selected library. This validated the parameters as valid proxies for improving the proteins. Each variant in the post-selection library was then scored and compared against initial filtering conditions to validate carbenicillin selection methodology.

Data from further rounds of carbenicillin selection is required to refine in-silico validation of tags.

4. Results

4.1 CAHS1 deep freeze-thaw selection

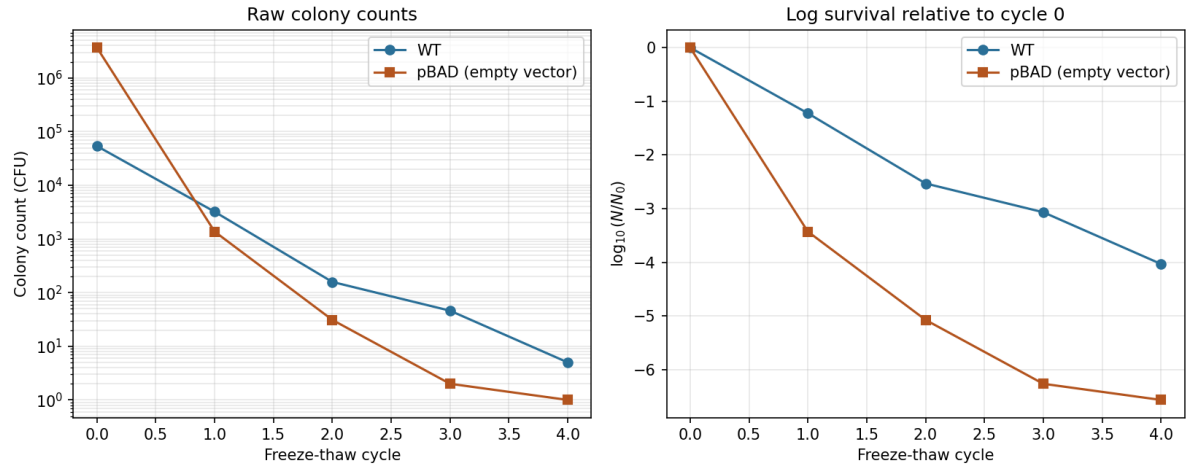


Fig. 4-1. Left panel: Change in raw pBAD and WT colony counts over freeze thaw cycles. Right panel: Log graph of cell survival across freeze-thaw cycles.

Cells expressing the wild-type CAHS1 sequence showed substantially higher survival across all freeze-thaw cycles than the empty-vector (pBAD) control, with the difference approaching approximately 2 to 3 orders of magnitude in log-survival by cycle 1 (Figure 4-1 right panel). This is consistent with CAHS1 conferring protection against freeze-thaw-induced cell death. This suggests that in principle, freeze-thaw can be used as a viable selection pressure to select for cells expressing favorable variants. Nonetheless, as this experiment was not replicated due to time and sample constraints, the result is merely preliminary and not statistically confirmed.

This experiment was performed with deep-freeze (slow) freezing with a -70°C freezer, whereas the library selection that would later yield results that would be sequenced used flash-freezing in isopropanol/dry ice. The result therefore does not directly confirm that the same protection applies under the selection conditions.

4.2 Library selection and sequencing

4.2.1 Culture OD600 change over time

OD600 readings of cell cultures containing untagged aggregation-prone β -lactamase and empty pBAD vector controls were observed to decrease after the addition of carbenicillin.

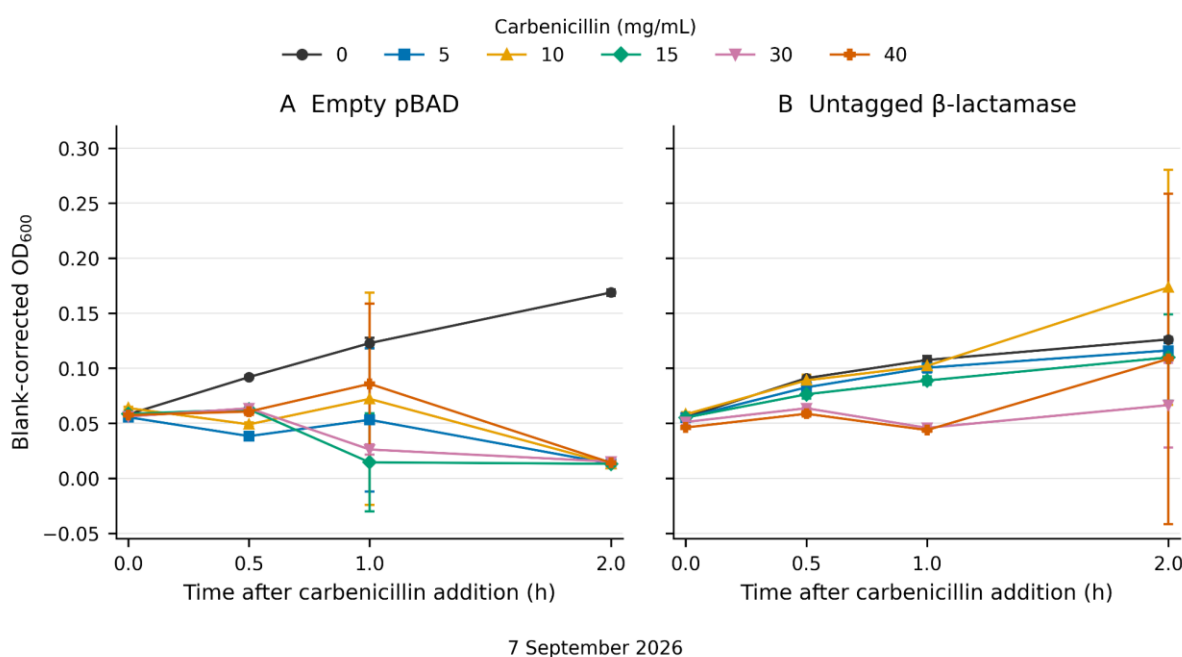


Figure 4-2. Blank corrected OD600 readings of empty pBAD vectors and untagged β -lactamase cell cultures over time. Carbenicillin was added at 0 hours together with 0.2% L-arabinose to induce β -lactamase production.

OD600 readings of cells containing empty pBAD vectors shows decrease in blank corrected OD600 from 0.058 to 0.001 over 2 hours after the addition of carbenicillin, as compared to cell

containing untagged β -lactamase showing increase in OD600 from 0.058 to 0.105-0.142 over 2 hours. This demonstrates that the carbenicillin selection is viable for benchmarking tag performance via rescuing β -lactamase, which can be quantified through change in OD600 readings over time under varying carbenicillin concentrations.

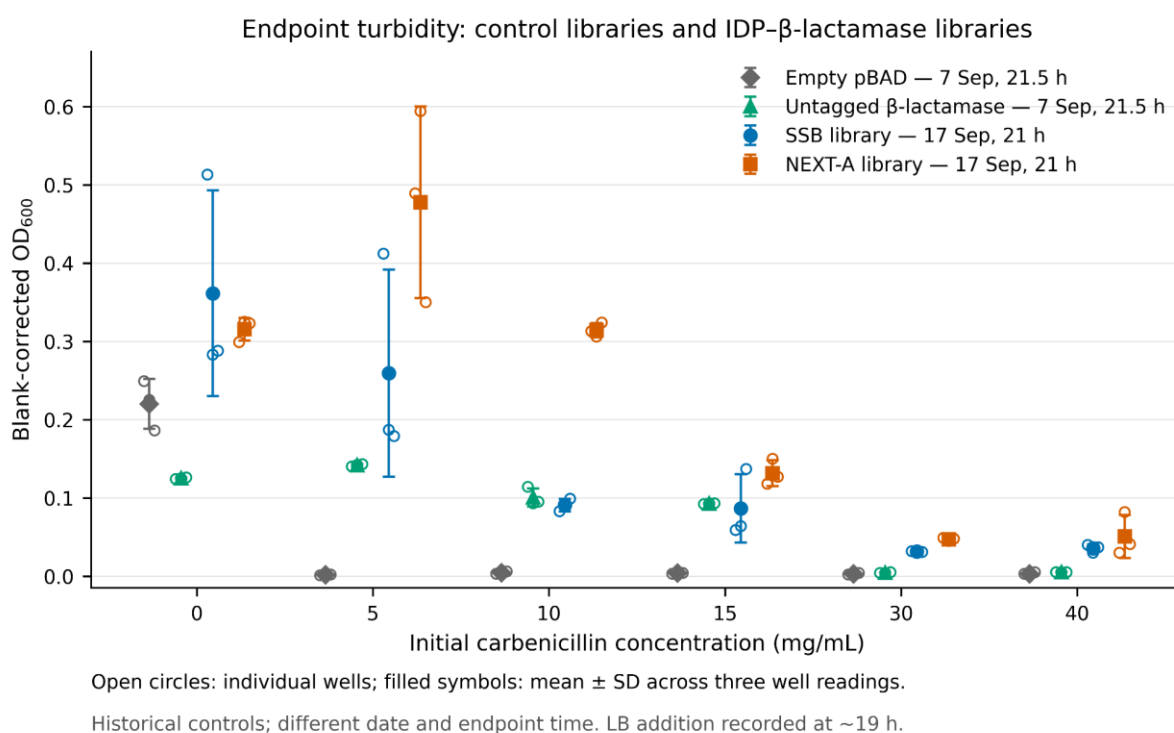


Figure 4-3. OD600 readings of cell cultures containing empty pBAD vectors, untagged β -lactamase, SSB and NEXT tagged β -lactamase at 21 hours of incubation. Cell cultures were incubated at 25°C with shaking. Figure generated from experimental OD600 data by OpenAI GPT-6 Astra.

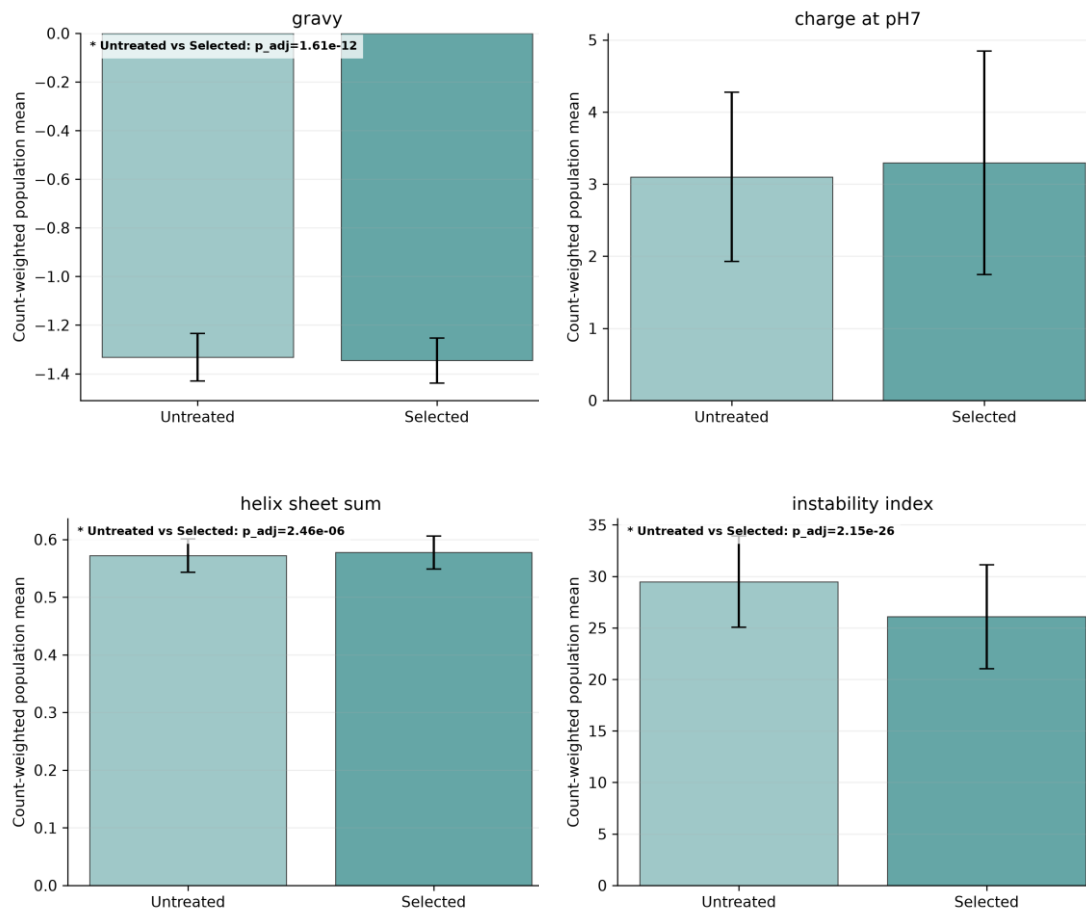
At 30,000-40,000 μ g/ml carbenicillin concentrations, endpoint values ranged from 0.018-0.059 compared to 0.291-0.361 for cultures without carbenicillin. At intermediate carbenicillin concentrations, NEXT derived tags reported higher OD readings of 0.314 at 10,000 μ g/ml carbenicillin, compared to 0.091 for SSB derived tags. Each reading for each tag at each carbenicillin concentration were taken as aliquots from the same culture and are only technical replicates.

OD600 readings were used as a proxy for cell survival based on the effect of cell lysis caused by carbenicillin selection on the population. The SSB and NEXT derived libraries can be shown to increase cell survival after 21 hours of incubation as shown by higher OD600 readings. Since readings are taken less than 24 hours after first addition of carbenicillin, we assume that the level of carbenicillin degradation is minimal and the endpoint OD600 readings are reflective of

the survivor count within each population.

4.2.2 NEXT sequencing data

The selected and untreated NEXT samples differed in amino-acid variant composition (Pearson $\chi^2 = 545.25$; fixed-margin Monte Carlo $p \approx 1 \times 10^{-5}$, at the simulation resolution limit). None of 100,000 simulated tables produced a statistic as large as observed, indicating that the difference was unlikely to arise from random read sampling alone under the equal-composition model.



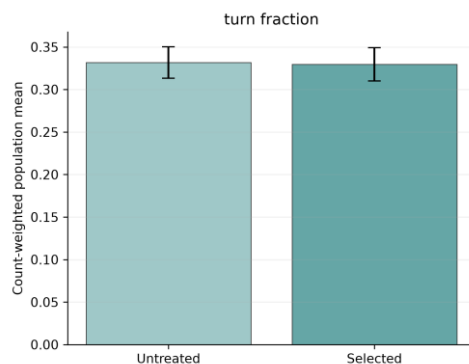


Figure 4-4. Comparison of GRAVY score, net charge at pH 7, helix sheet sum, instability index and turn fraction of NEXT derived library before and post selection. * Indicates statistical significance by Mann-Whitney U Test.

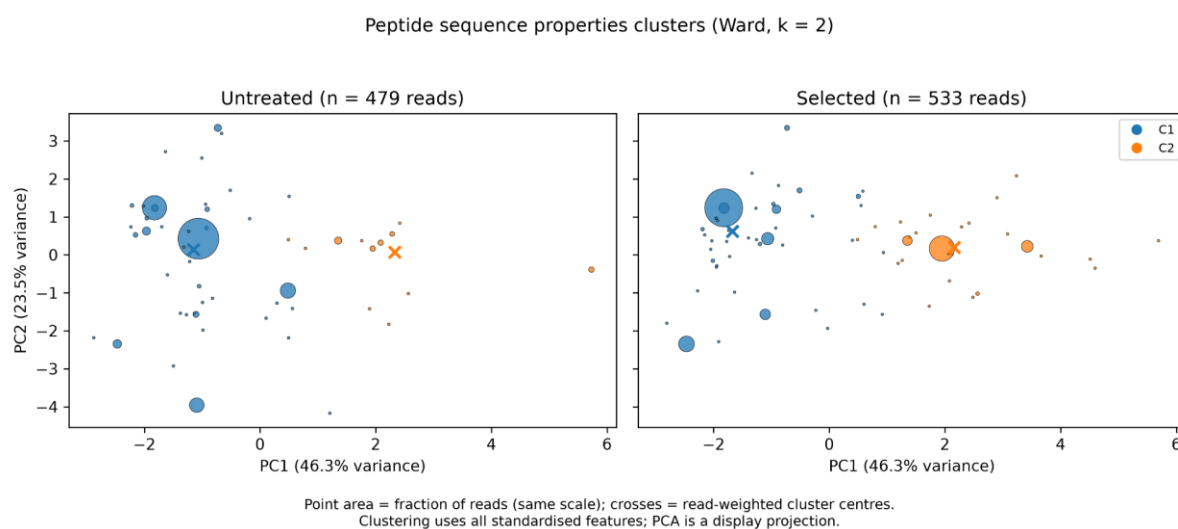


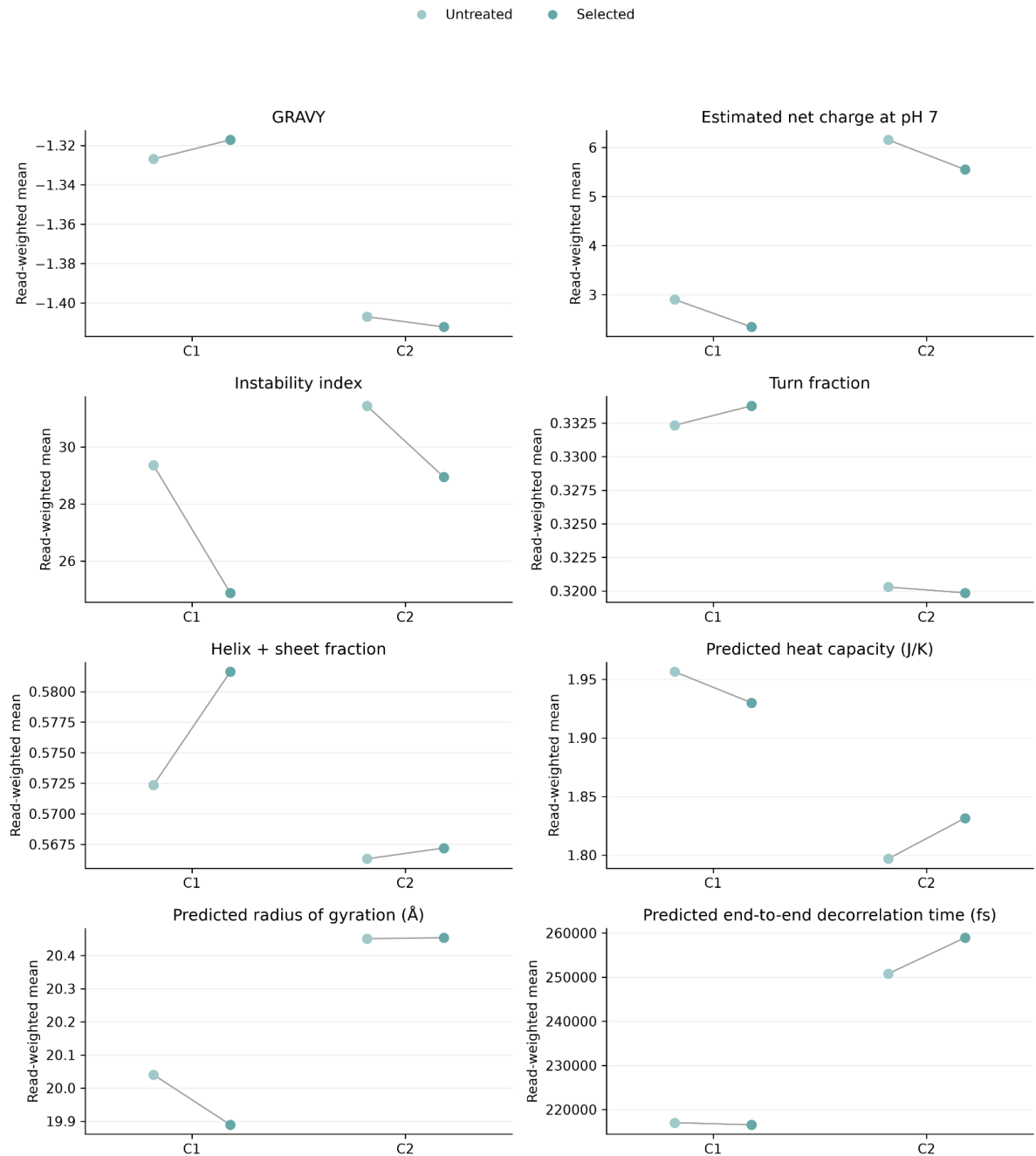
Figure 4-5. Cluster diagrams of unrated and selected NEXT-derived libraries.

Table 4-1. Proportional population size of different clusters before and after selection in NEXT library.

	C1 proportion	C2 proportion
Untreated	93.9%	6.1%
selected	70.4%	29.6%

Cluster C2 shows enrichment post selection, increasing approximately 4.8-fold in relative abundance, while the proportions assigned to C1 decreased.

B. NEXT: traits within each fixed cluster



Each point is a within-cluster read-weighted mean; connecting lines show changes across samples.

Figure 4-6. Comparison of read-weighted means of protein properties between clusters before and after selection.

In the selected sample, C2 had a higher read-weighted mean predicted net charge (5.55) than C1 (2.34), as well as a lower GRAVY score of (-1.41) compared to C1 (-1.32). The mean instability index decreased within both clusters, with the largest reduction observed in C1, from 29.4 to 24.9. The combined helix- and sheet-associated residue fractions of C2 remained broadly similar before and after selection. Post selection, C2 has a higher mean predicted radius of gyration (20.5) as compared to C1 (19.5), as well as higher end-to-end correlation time (258947fs) compared to C1 (216561fs).

4.2.3 SSB sequencing data

The selected and untreated SSB samples differed in amino-acid variant composition (Pearson $\chi^2 = 1026.1$; fixed-margin Monte Carlo $p \approx 1 \times 10^{-5}$, at the simulation resolution limit). None of 100,000 simulated tables produced a statistic as large as observed, indicating that the difference was unlikely to arise from random read sampling alone under the equal-composition model.

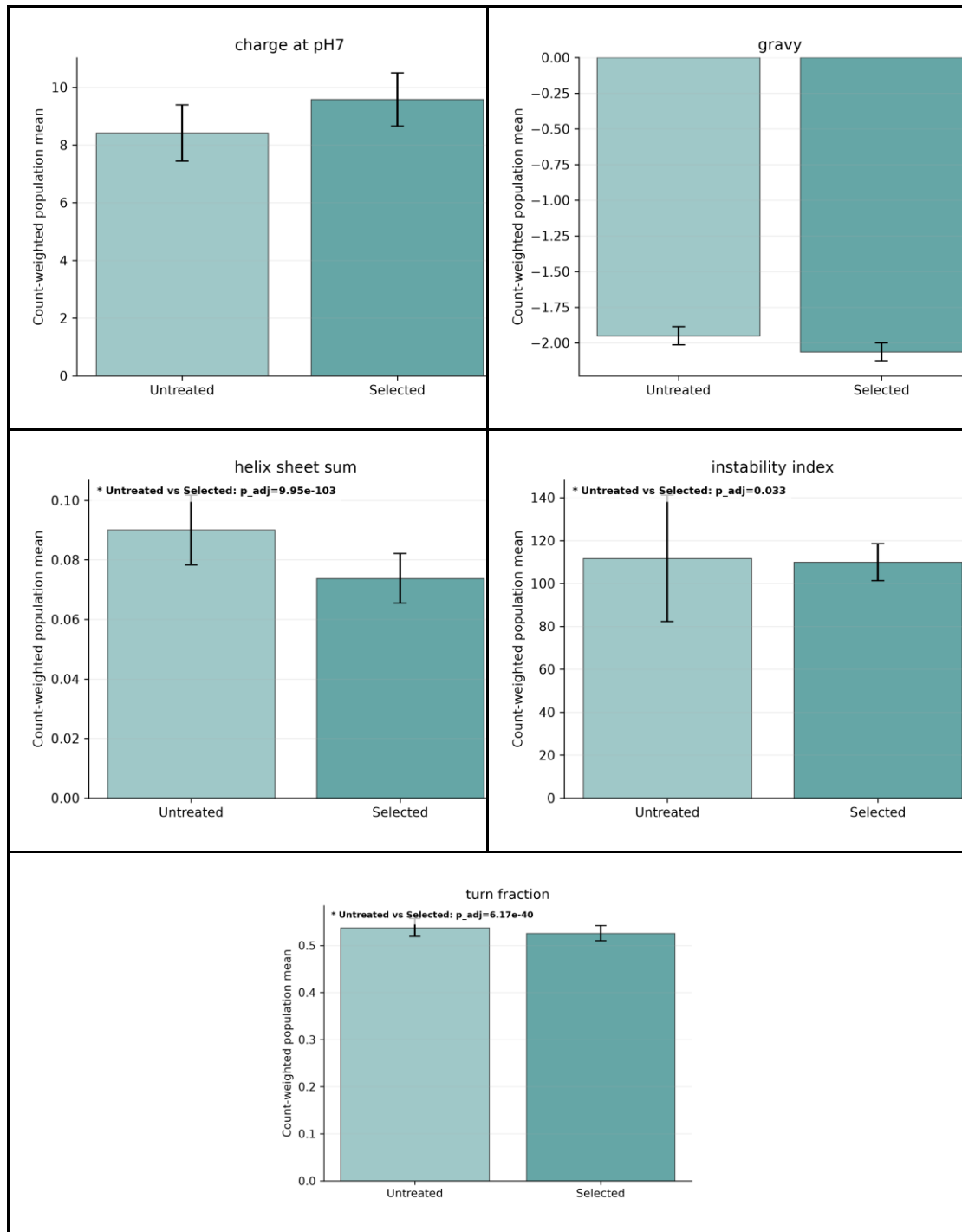


Figure 4-7. Comparison of GRAVY score, net charge at pH 7, helix sheet sum, instability index and turn fraction of SSB derived library before and post selection. * Indicates statistical significance by Mann-Whitney U Test.

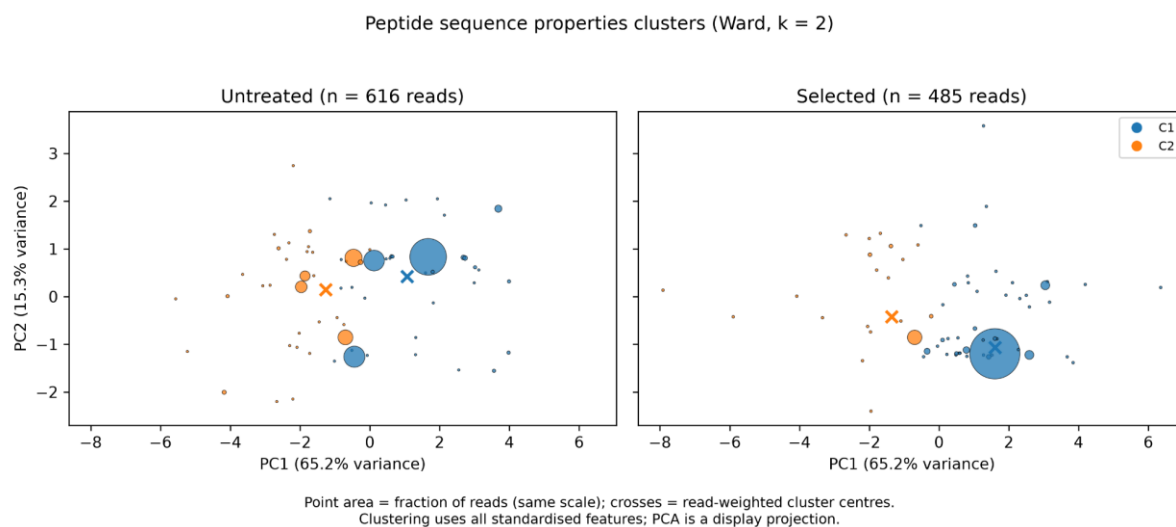


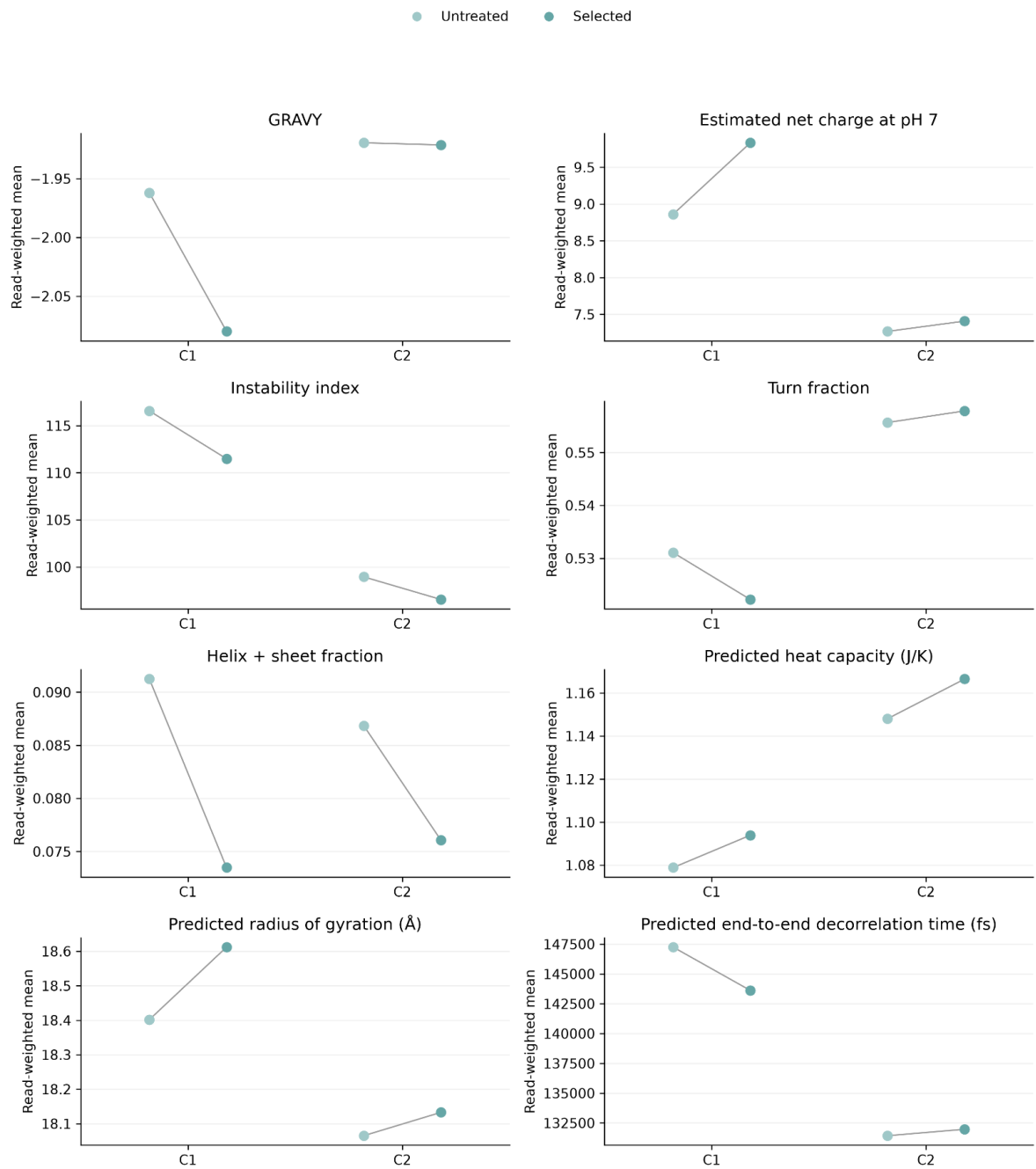
Figure 4-8. Cluster diagrams of unrated and selected SSB-derived libraries.

Table 4-2. Proportional population size of different clusters before and after selection in SSB library.

	C1 proportion	C2 proportion
Untreated	72.2%	27.8%
Selected	89.5%	10.5%

Hierarchical clustering of the pooled untreated and selected variants produced two clusters, with a silhouette score of 0.321. The selected sample contained a greater proportion of reads assigned to C1. Much of this increase was attributable to SSB-054, which accounted for 352 reads (72.6%) in the selected sample and was not detected in the untreated sample.

B. SSB: traits within each fixed cluster



Each point is a within-cluster read-weighted mean; connecting lines show changes across samples.

Figure 4-9. Comparison of read-weighted means of IDP properties between clusters before and after selection.

Within C1, the read-weighted mean GRAVY decreased from -1.96 to -2.08 , while predicted net charge increased from 8.86 to 9.83 . Across the both clusters, there are decreases in read-weighted mean instability index and combined helix- and sheet-associated residue fractions. A higher read-averaged mean predicted radius of gyration ($18.61\text{\AA}$) and end-to-end decorrelation time (143617fs) was observed in C1 after selection.

4.2.4 De novo sequencing data

Sequencing of the De Novo library revealed significant contamination from the NEXT library that had likely occurred during cloning. The OD readings and sequence data for the De Novo library will not be discussed further in this report.

4.2.5 CAHS1 sequencing data

CAHS1 samples differed in amino-acid variant composition between 0, 3 and 6 rounds of freeze-thaw selection (Pearson $\chi^2 = 637.8, 286.5$; fixed-margin Monte Carlo $p \approx 1 \times 10^{-5}$, at the simulation resolution limit). None of 100,000 simulated tables for each pair of samples produced a statistic as large as observed, indicating that the difference was unlikely to arise from random read sampling alone under the equal-composition model.

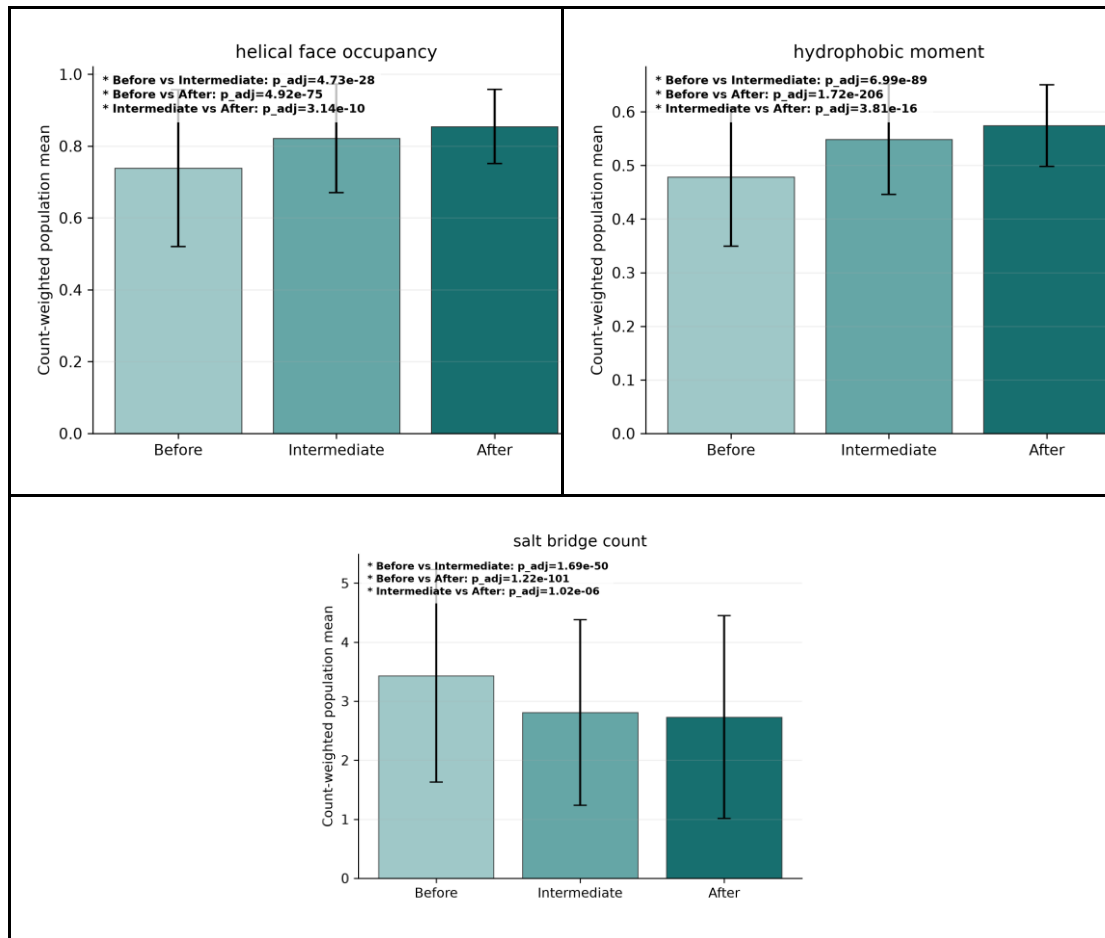


Figure 4-10. Comparison of helical face occupancy, hydrophobic moment and salt bridge counts of CAHS1 derived library at 0 rounds (Before), 3 rounds (Intermediate) and 6 rounds (After) of freeze-thaw selection.. *

Indicates statistical significance by Mann-Whitney U Test.

Clustering was performed on sequencing data for samples that had gone through 0, 3 and 6 freeze thaw cycles. Across selection rounds, decreases in Pace-Scholtz sums and means as well as increase in helical face occupancy were observed. The cluster maps and traits within each cluster are shown below.

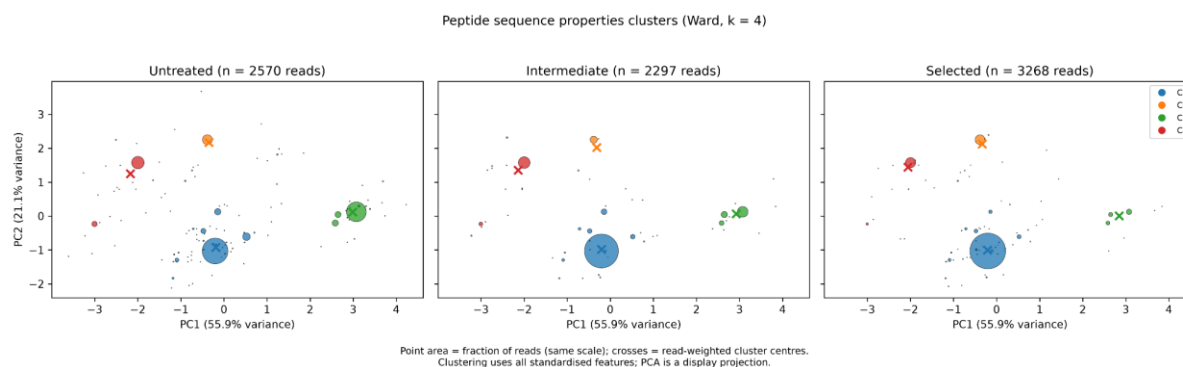


Figure 4-11. Cluster diagrams of CAHS1-derived libraries at 0 (untreated), 3 (intermediate) and 6 (selected) rounds of freeze-thaw selection.

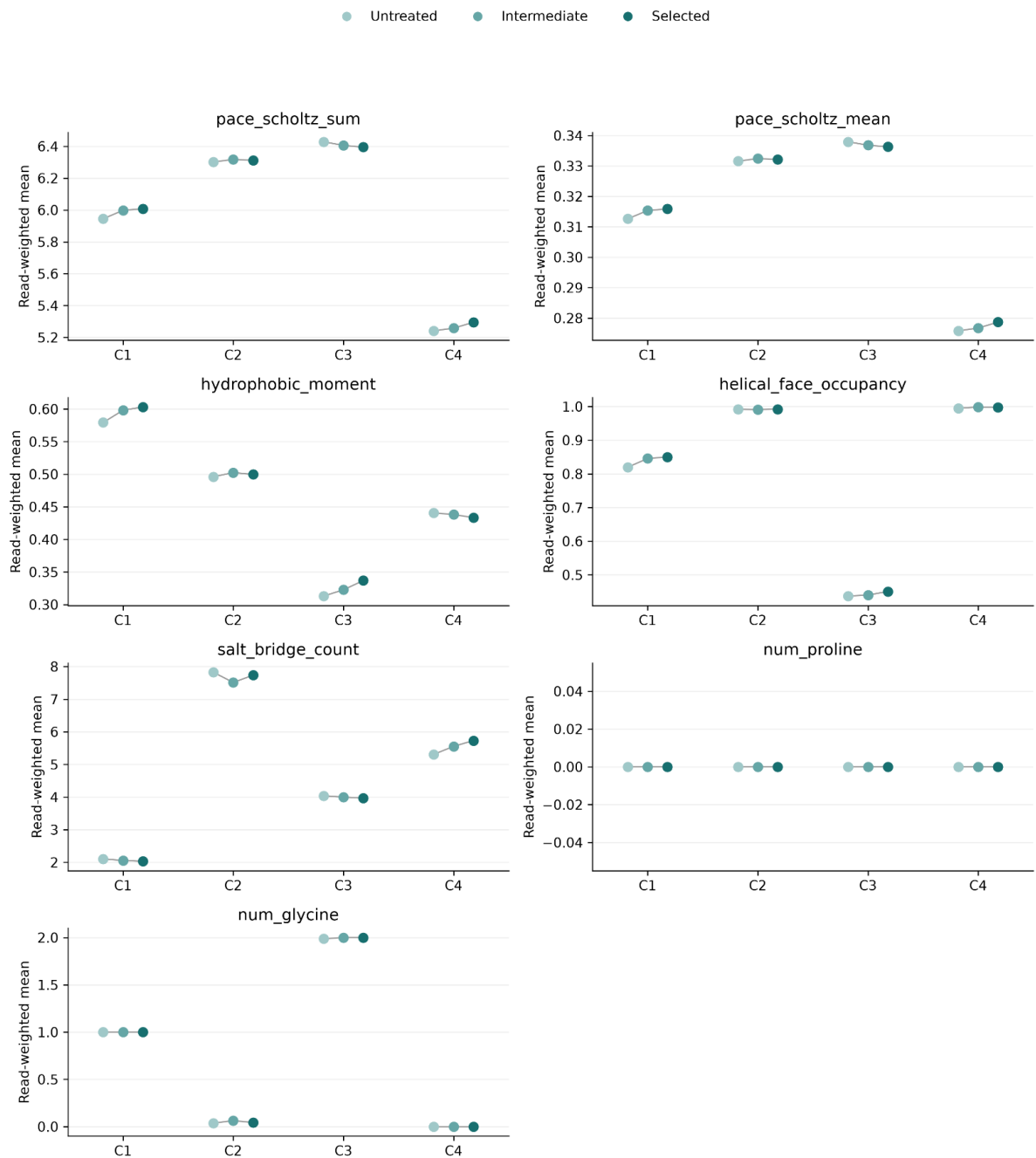
Table 4-3. Proportional population size of different clusters in the library of CAHS before, during, and after selection.

	C1 proportion	C2 proportion	C3 proportion	C4 proportion
Round 0	51.9%	6.38%	29.6%	12.1%
Round 3	75.6%	3.48%	10.9%	10.0%
Round 6	82.7%	6.43%	3.79%	7.04%

In the current four-cluster representation (mean silhouette score = 0.367), the increased representation of C1 was accompanied by decreased representation of C3 and C4. The read-weighted within-cluster PCA centres remained comparatively similar, indicating that substantial changes in variant frequencies were accompanied by smaller changes in the average properties represented by the first two principal components.

The dominant variant identified by enrichment in C1, CAHS1-055, increased from 1,027 of 2,570 reads (40.0%) in the round 0 sample to 2,509 of 3,268 reads (76.8%) after round 6. This represented an approximately 1.92-fold increase in relative abundance.

B. CAHS1: traits within each fixed cluster



Each point is a within-cluster read-weighted mean; connecting lines show changes across samples.

Figure 4-12. Comparison of read-weighted means of CAHS1 related properties between clusters before and after selection.

Post selection, C1 displayed intermediate values of read-weighted mean Pace-Scholtz sum (6.01) and mean (0.316). C1 displayed the lowest read-weighted mean salt bridge count (2.03) compared to C3 (3.97) and C4 (5.73). C2 counts for the lowest representation among all clusters and displayed no significant change in percentage representation within the population across all 3 rounds and displayed the highest salt bridge count post selection (7.73).

4.2.6 Error-prone PCR of NEXT tag

3 consecutive rounds of error prone PCR (epPCR) were performed on the original NEXT tag to serve as a baseline consisting of random nucleotide substitutions to compare against selected samples. Cloning and expression of the epPCR derived library was not performed. A PCA plot of the epPCR NEXT population was plotted. NEXT-072, a candidate identified by enrichment during selection of NEXT-derived tags, is plotted as reference.

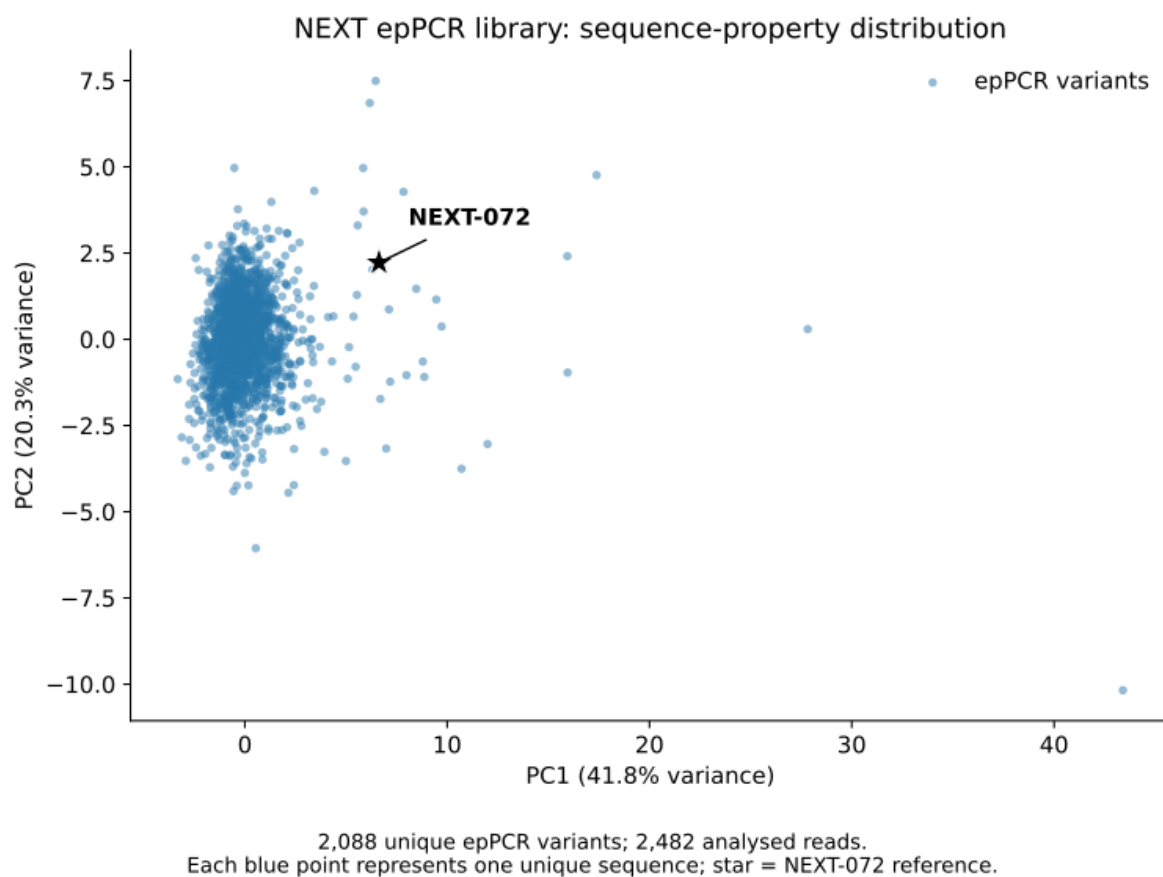


Figure 4-13. PCA plot of epPCR mutants of NEXT tag. Mutants did not undergo the same selection procedures. NEXT-072 derived from experimental selection included for reference. Figure generated from sequencing data by

OpenAI GPT-6 Astra.

Over 3-4 repeated rounds of epPCR, mutated template seems unable to be amplified further with sufficient yield. Therefore, we propose that this current library of NEXT IDP reached nearly maximum mutation rate (5.5 mutations per 200 bp per three rounds of epPCR) achievable by epPCR with current methodology. However, as shown in Figure 13, the most successful NEXT mutant from our computationally designed library has different characteristics compared to most epPCR mutants. Therefore, it is likely that simply using epPCR would not have produced mutant IDPs with sufficiently diverse characteristics for selection. This highlights the necessity of performing large scale in-silico mutagenesis in constructing the mutant library for IDP selection. However, PCA position does not quantify the probability of generating a particular sequence. To establish the advantage of in-silico mutagenesis over epPCR would require comparable libraries subjected to the same selection and IDP functional validation.

4.3 Validation

4.3.2 Solubility assay of commercially available tags

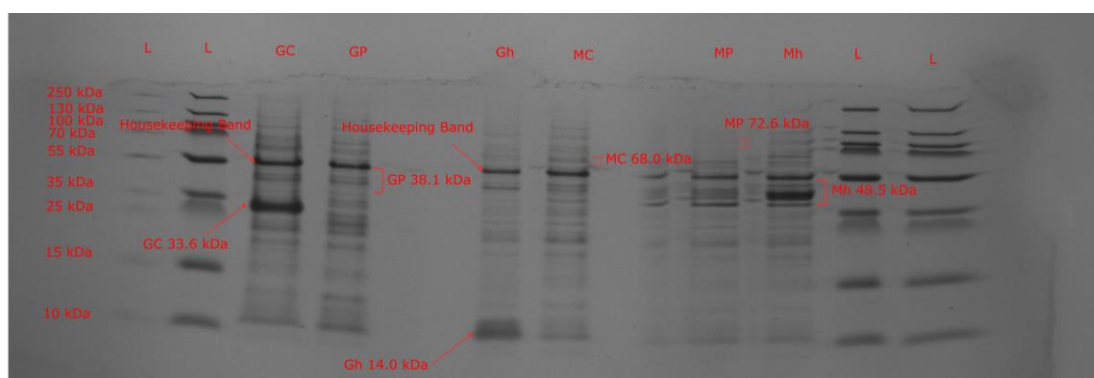


Figure 4-14. Second elution samples across six constructs for densitometric analysis. From left to right: GB1-taCA fusion, GB1-isPETase fusion, GB1-hEGF fusion, MBP-taCA fusion, MBP-isPETase fusion, MBP-hEGF fusion.

Identified proteins were labeled directly. Unidentifiable proteins had expected region appearance labelled in brackets. Housekeeping bands have been labelled for signal normalization.

Densitometric analysis was only performed on GB1-taCA and GB1-hEGF fusions due to unsuccessful purification or assembly of other constructs. GB1-taCA and GB1-hEGF have their associated normalised signal (0.7459 & 0.6856) which can be used to compare against other strains to quantify improvement in performance of the tags. Further work has to be done to quantify solubility improvements conferred by tag fusions.

5. Discussion

5.1 Sequencing

The changes in NEXT and SSB libraries favour clusters with more negative GRAVY values and higher net charge, consistent with the hypothesis that hydrophilic, charged sequences support solubility. Future work may focus on these factors as design choices when considering solubility tag sequences.

Decreases in instability indices were observed in SSB and NEXT tags post selection. Further work is needed to verify if this is due to the C terminus placement of IDP tags leading to a selection against C-degron motifs.

For CAHS1, the current analysis illustrates how substantial sequence-level redistribution can coexist with comparatively stable within-cluster PCA centres: variants can differ in abundance while occupying similar regions of the measured property space.

From statistical analysis, it is shown that on a population level, the salt-bridge count decreases, and on a cluster-level, C1 has the lowest salt-bridge count while being the most enriched. These results suggest that this allows for greater flexibility or fewer charge-mediated intermolecular contacts that promote aggregation. There is also an increase in helical face occupancy and hydrophobic moment, consistent with enrichment of variants with stronger amphipathic helical character.

The dominance of individual variants highlights candidates for follow-up testing but may also make cluster-level trends heavily dependent on a single sequence. Repeating selection with independent cultures and testing reconstructed variants against their parental tags would help establish whether enrichment corresponds to improved function. Soluble protein yield, β -lactamase activity, and freeze-thaw survival measurements would then connect the sequencing observations to the intended applications.

5.3 Validation Assays

5.3.1 Functional Assay

The β -lactamase selection system partially validates non-disruptive impact on enzyme function as it was able to rescue the β -lactamase indicated by higher OD600 readings in 30k and 40k $\mu\text{g/mL}$ concentrations.

Functional assays to quantify change in enzymatic activity were planned but failed to be carried out due to time constraints. taCA and isPETase were chosen targets due to the ability to act as an esterase. Enzyme activity was to be characterised through colorimetric assays to validate the impact of the tag on the enzyme function.

5.3.2 Solubility Assay

Solubility assay was carried out via densitometric analysis. Due to time and budget constraints, the solubility assay was not carried out in completion and insufficient data to conclude.

5.3.3 In Silico Validation

Wet lab results suggest that parameters used in in-silico selection are good proxies for protein performance. It is unknown whether the change in parameters from pre to post selection libraries is optimal. Further rounds of selection will inform whether parameters are optimal or whether the optimal is much further away. A refinement of the model is to find the optimal range by repeating selection rounds until parameters no longer fluctuate. This will then produce a model that can directly score the performance of each tag based on this range such that successful tags can be differentiated based on deviation from optimal parameter value. Further work could use the in-silico model to benchmark the performance of the epPCR generated library which could be assembled and selected to further validate the in-silico model.

6. Conclusion

After wet lab experiments and statistical analysis on the sequencing data, these are the enriched tag sequences from each tag family. These tags showed the largest increase in proportion within the population's post selection and show improvement consistent with the internal in-silico model produced.

NEXT-072:

Amino acid sequence:

AVQHSNGNLSDLGAEMKKQHKTKNPEGHASAQGKAGAHNRKKKEEAPKPKPVV

SSB-054:

Amino acid sequence:

RQGGGNRGRSNIGGGQPQGGKRRPQQRQGGNQFSGGRGRRPQ

CAHS1-055:

Amino acid sequence:

IRRQMRRRAKGRIRKEMTKM

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Generative AI tools have been used in this project for experiment brainstorming, literature research, as well as English language writing, generation of selected schematics and the analysis and debugging of code. All scientific data outputs, references and code were

reviewed by the team before inclusion into the project. Reasonable efforts have been made where possible to verify claims from generative AI tools.